Arithmetic structure of the standard genetic code

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Abstract

We analyze proton and neutron counts of free neutral amino acids across 33 codon groups of the standard genetic code, defined by two natural grouping systems: the Rumer partition and the three third-position chemical axes—Keto/Amino, Strong/Weak, and Purine/Pyrimidine. After excluding the three stop codons and the initiator ATG, the neutron sum of the remaining 60 codons is divisible by 37, while the proton sum is one below the nearest multiple. The same pattern appears in the Keto/Amino asymmetry, which reduces to the tryptophan–isoleucine pair with neutron and proton differences of 37 and 36. Proton parity excludes a difference of 37 but does not determine the one-unit deviation. To analyze the symmetries of the complete table, we assign equal formal contributions to the three stop codons and a separate one to the initiator. Keto/Amino balance and divisibility of the global proton and neutron sums by 37 yield a unique minimal non-negative integer solution. The three stop codons contribute $111 = 3 \cdot 37$ neutrons and the initiator none, equating the global neutron sum with the Octet I nucleon sum without imposing this equality. Across 27 NCBI tables, the neutron deficit relative to the Octet I nucleon sum and the proton residue recur only with the same service-excluded pool. Controls highlight third-position groupings and 37 under chosen measures. Monte Carlo comparison with random permutations of amino acids among synonymous blocks shows an atypical proximity of the standard-code group sums to multiples of 37; the result persists when methionine is held fixed. These group-sum patterns depend jointly on codon multiplicity and nucleon composition.

Contents

1 Introduction **2**

2 Methods **3**
2.1 Nucleon composition of amino acids . . . 3
2.2 Genetic code structure . . . 4
2.3 Codon-group construction . . . 6
2.4 Pyrimidine synonymy and twin-group identities . . . 7
2.5 Parametrization of service codons . . . 8
2.6 Analytical criteria and classification of equalities . . . 9
2.7 Computed datasets and reproducibility . . . 11

3 Results **13**
3.1 Parametrization-independent results . . . 14
3.1.1 The service-excluded pool . . . 14
3.1.2 Arithmetic of Octet I . . . 15
3.1.3 Arithmetic of pyrimidine-restricted groups . . . 17
3.2 Results under Key 0 . . . 18
3.2.1 Divisibility of T by 37 at the upper levels of partitioning . . . 20
3.2.2 Alignment of Amino and Weak with the lattice of step 37 . . . 20

3.2.3 One-unit deviations in Δ and the {G} branch . . . 21
3.3 Localization of the Keto/Amino asymmetry and proton parity . . . 21
3.3.1 Localization of the Keto/Amino asymmetry . . . 22
3.3.2 Proton parity . . . 23
3.4 Analytical derivation of Key 1 . . . 24
3.4.1 Determination of the neutron parameters . . . 24
3.4.2 Determination of the proton parameters . . . 25
3.4.3 Hierarchy of constraints and the choice of Key 1 . . . 27
3.5 Consequences under Key 1 . . . 29
3.5.1 Alignment of the chemical-axis 32-codon groups . . . 29
3.5.2 Macroscopic identities and propagation to chemical-axis halves . . . 29
3.5.3 Divisibility by 37 and by 999 . . . 31
3.5.4 The Δ-invariant across all six 32-codon groups . . . 32
3.5.5 Purine/Pyrimidine asymmetry . . . 32
3.6 Synthesis . . . 33

4 Discussion **35**
4.1 Specificity across alternative genetic codes . . . 35
4.2 Specificity of the third codon position, the modulus 37, and the molecular representation . . . 37
4.3 Significance under a random-code null . . . 40
4.4 Relation to prior work . . . 42
4.5 Limitations and outlook . . . 43

5 Conclusion **44**

1 Introduction

The genetic code is one of the most ancient and conserved molecular features of cellular organisms. Its near-complete universality across the known domains of life, and the limited scope of variation in the known alternative codes, indicate that the basic structure of the standard code was already established at the early stages of biological evolution. Any organizational regularities present in its arithmetic structure therefore require precise analysis as one of the few potential sources of information about the early stages of life.

The arithmetic structure of the standard genetic code has been studied since shortly after its decipherment. Yu. B. Rumer [6, 7] introduced the partition of the 64 codons into two classes of 32 codons each, Octet I (the eight four-fold degenerate families) and Octet II (the remaining codons), related by the involution $R = (\mathrm{T} \leftrightarrow \mathrm{G}, \mathrm{C} \leftrightarrow \mathrm{A})$. Shcherbak and Makukov [8] observed that the number 37 plays a recurrent role in the arithmetic of the code, appearing as a divisor of amino acid nucleon sums across various groupings of codons. Panov and Filatov [5] revisited and extended this analysis, using per-codon counting, which forms the methodological basis of the present study.

The present analysis uses the proton and neutron counts of the twenty canonical amino acids in their free neutral forms under the most-abundant-isotope convention; once these conventions are fixed, the counts are uniquely determined. These quantities are considered together with two natural grouping systems defined independently of the nucleon arithmetic. The first is the Rumer partition. The second is given by the three chemical axes of the third codon position — Keto/Amino, Strong/Weak, and Purine/Pyrimidine — which exhaust the partitions of the four nucleotide bases into two pairs and correspond to the functional-group, hydrogen-bond, and ring-topology dichotomies of nucleobase chemistry.

In the present work, the arithmetic regularities of the standard genetic code are examined within this single grouping system. We identify the initial numerical relations and trace the equalities and divisibilities that follow from them across different parts of the codon table.

To assess how specific these regularities are to the standard code, we compare it with alternative NCBI codes and use Monte Carlo simulations to estimate how often comparable results arise under random permutations of amino acids among the synonymous blocks. We also examine whether the third codon position stands out among the three positions and the number 37 among the other prime moduli, and how the results depend on the molecular form used to represent the amino acids.

The analysis is divided into three levels. The principal regularities are established at a parametrization-independent level, at which the three stop codons and ATG in its initiator role are excluded and only the remaining 60 codons are summed; these are the central results of the paper, and they involve no assigned values. To obtain sums for the complete table of 64 codons, however, the four service codons must be given some nucleon contributions. The second level is the naive reading (Key 0), in which the stop codons are assigned zero and ATG is counted as methionine. Assigning zero is also a choice of value, not the absence of one. The third level is the symmetrization (Key 1), in which the four service codons carry derived nucleon contributions. Key 1 does not establish the parametrization-independent core of the structure: that core is already present in sums over the 60 codons remaining after the service codons are excluded. The role of Key 1 is to show how the structure extends to the complete table, taking into account the fixed distribution of the service codons across the groups in the standard code. For the chosen symmetry conditions, these contributions are determined as the unique minimal non-negative integer solution of the corresponding system (Section 3), and the resulting regularities are consequences of that solution.

Assigning values to the service codons is a formal device for treating the arithmetic of the complete table as a single object; neither the assigned values nor the sums that include them are given a physical interpretation.

2 Methods

This section defines the nucleon parameters, the codon group systematics, and the parametrizations of service codons used throughout the analysis, together with the computed datasets supporting it.

2.1 Nucleon composition of amino acids

Throughout this work, each of the twenty canonical amino acids is represented as a free neutral amino acid molecule (not as a residue in a peptide chain), with its molecular formula retrieved from the PubChem compound database [3]. From this formula, we compute four integer quantities that characterize the nucleon content of the molecule. Below, $Q(a)$ denotes any of the four quantities T, P, N, or Δ for one molecule of amino acid a. The sensitivity of the results to this choice of molecular representation is examined in Section 4.2.

The proton count P is defined from the elemental composition of the molecular formula as

$$P = \sum_E n_E Z_E,$$

where n_E is the number of atoms of element E in the formula and Z_E is its atomic number. The atomic numbers used are $Z(\mathrm{H}) = 1$, $Z(\mathrm{C}) = 6$, $Z(\mathrm{N}) = 7$, $Z(\mathrm{O}) = 8$ and $Z(\mathrm{S}) = 16$.

The neutron count N is computed by assigning each element its most abundant naturally occurring stable isotope, using the terrestrial isotopic compositions evaluated by IUPAC [1].

Accordingly, we use ^{1}H, ^{12}C, ^{14}N, ^{16}O and ^{32}S. The total neutron count is

$$N = \sum_{E} n_E (A_E - Z_E),$$

where A_E is the mass number of the selected isotope of element E.

The total nucleon count T and the proton–neutron difference Δ are defined as

$$T = P + N, \qquad \Delta = P - N.$$

Both quantities are integers by construction. Under the isotope assumptions above, hydrogen is the only element contributing to Δ, since all other isotopes used have equal numbers of protons and neutrons. Consequently, Δ is numerically equal to the number of hydrogen atoms in the molecule and, in particular, $\Delta \geq 0$ for each of the twenty canonical amino acids. By linearity, the same identity holds for any sum of canonical amino-acid contributions and, in particular, for all group sums over the service-excluded pool (Section 2.2).

To avoid a notational collision, we use uppercase Δ exclusively for the proton–neutron imbalance of a single molecule or codon group, and reserve lowercase δ for pairwise differences between the molecular quantities of two amino acids (for example, $\delta P = P_i - P_j$ between two amino acids i and j). When both amino acids are identified explicitly, their labels are placed in parentheses:

$$\delta Q(a, b) = Q(a) - Q(b).$$

We emphasize that P, N, T and Δ are not measured molecular masses. They are exact integer quantities derived from elemental composition, and all arithmetic identities reported in this study are exact integer relations rather than artifacts of rounding approximate molecular masses.

The nucleon values for all twenty canonical amino acids are collected in the input dataset `amino_acids_nucleons.csv`, with one row per amino acid and columns for P, N, T and Δ. From this table we compute three companion datasets of pairwise absolute differences: `amino_acids_proton_differences.csv`, `amino_acids_neutron_differences.csv` and `amino_acids_nucleon_differences.csv`. For any pair of amino acids (i, j) we write $\delta P = P_i - P_j$ and analogously for N and T. Each file groups all $\binom{20}{2} = 190$ unordered pairs (i, j) by the corresponding absolute difference $|\delta P(i, j)|$, $|\delta N(i, j)|$, or $|\delta T(i, j)|$.

All subsequent nucleon computations for codon groups reduce, by linearity, to sums of the per-amino-acid values tabulated in `amino_acids_nucleons.csv`, weighted by the multiplicities of the corresponding codons. Where service codons are included, their contributions are determined by the chosen parametrization (Section 2.5).

2.2 Genetic code structure

The standard genetic code maps the 64 codons of the four-letter nucleotide alphabet $\{\mathrm{A}, \mathrm{T}, \mathrm{G}, \mathrm{C}\}$ onto the twenty canonical amino acids and a translation-termination signal. Throughout this paper, we use DNA notation (with thymine T rather than uracil U) and adopt the codon assignments of the standard code (NCBI translation table 1) [4]. The three codons TAA, TAG and TGA act as stop signals, whereas ATG encodes methionine and serves as the canonical translation-initiation signal. For the purposes of the present arithmetic analysis, these four codons — the three stop codons and ATG in its initiator role — are referred to as *service codons*. Among possible initiation codons, ATG is predominant. Its RNA equivalent AUG is the only one of them fully complementary to the CAU anticodon of the initiator methionine tRNA. In the present analysis, ATG represents the initiation function; this analytical convention does not exclude initiation from other codons.

Let the complete set of 64 codons be

$$\mathcal{C}_{64} = \{\mathrm{A}, \mathrm{T}, \mathrm{G}, \mathrm{C}\}^3,$$

and let the four service codons form the set

$$\mathcal{S} = \{\mathrm{ATG}, \mathrm{TAA}, \mathrm{TAG}, \mathrm{TGA}\}.$$

The *service-codon-excluded 60-codon pool* is then defined as

$$\mathcal{C}_{\mathrm{SE}} = \mathcal{C}_{64} \setminus \mathcal{S},$$

with $|\mathcal{C}_{\mathrm{SE}}| = 60$. Every codon belonging to $\mathcal{C}_{\mathrm{SE}}$ is counted according to its standard amino-acid assignment. Because ATG is the sole methionine codon, $\mathcal{C}_{\mathrm{SE}}$ contains codons for nineteen canonical amino acids; methionine is absent from this pool. The full codon-to-product mapping is provided in the dataset `genetic_code_codons.csv`.

By a codon family, we mean the set of four codons sharing the same first two bases and having an arbitrary third-position base, denoted by N. A structural property of the standard code, first identified by Rumer [6, 7], is the partition of the 64 codons into two halves of 32 codons each, distinguished by the degeneracy pattern of the third codon position. The first half, which we denote *Octet I*, consists of the eight codon families in which all four third-position nucleotides encode the same amino acid. These fully four-fold degenerate families are GCN (Ala), CGN (Arg), GGN (Gly), CCN (Pro), ACN (Thr), GTN (Val), CTN (Leu) and TCN (Ser). The second half, denoted *Octet II*, consists of the remaining eight codon families, in which the third-position nucleotide affects the encoded product. Every service codon of the standard code belongs to Octet II and carries a purine (A or G) at the third position.

The Rumer partition is not an analytical construct introduced in the present work but a structural feature of the code itself. It reflects both the qualitative distinction between fully and partially degenerate codon families and the quantitative fact that these two classes have equal cardinality: each contains eight families and thirty-two codons. Because Octet I contains no service codons, all nucleon quantities computed over Octet I are independent of any parametrization of start and stop signals and are determined solely by the nucleon content of the encoded amino acids. The same property holds, for a separate reason, for any codon group whose third-position nucleotide is restricted to pyrimidines (cytosine or thymine), since all four service codons of the standard code carry a purine at the third position. Full four-fold degeneracy and a purine-free third position are two distinct structural conditions, but both have the same consequence: nucleon quantities over the corresponding groups are fixed by their amino-acid composition and provide parametrization-independent reference values for the arithmetic identities of Section 3.

A further structural property of the Rumer partition, introduced by Rumer [7], is that the two octets are related by a simple permutation of the nucleotide alphabet. Under the transformation

$$R = (\mathrm{T} \leftrightarrow \mathrm{G},\ \mathrm{C} \leftrightarrow \mathrm{A}),$$

each codon family of Octet I is mapped to a codon family of Octet II, and vice versa. This transformation, known as the Rumer transformation, rearranges the partition but does not preserve the amino acids encoded by individual codons.

Taken together, the four-fold degeneracy pattern and the Rumer transformation identify the Octet I / Octet II partition as a natural structural division of the standard code, independent of the nucleon arithmetic. In the present work, this division is used as a reference partition over which nucleon quantities are computed and compared.

The codon table (Figure 1) is displayed with its axes in the base ordering C, G, T, A, which is also the default ordering of the accompanying visualizer. This ordering has two justifications. Chemically, it was introduced by Rumer [7] on the basis of base type and the number of hydrogen bonds in the canonical base pair. The sequence C, G, T, A alternates pyrimidines and purines, while within each of these classes the member of the Strong pair $\{\mathrm{C}, \mathrm{G}\}$ precedes the member of the Weak pair $\{\mathrm{T}, \mathrm{A}\}$: C precedes T among the pyrimidines, and G precedes A among the purines. Geometrically, as shown by Panov and Filatov [5], it is one of exactly four orderings

(out of 24 possible permutations of the four bases) in which Octet I and Octet II form connected regions on the codon matrix.

To assess the extent to which the arithmetic properties studied in this paper are specific to the standard genetic code, we additionally retrieved the full set of 27 translation tables maintained by NCBI [4]. The tables are numbered 1–33, with some numbers left unused, and cover nuclear, mitochondrial and plastid codes across a broad taxonomic range. The codon assignments of all these tables are collected in the dataset `ncbi_genetic_code_registry.csv` and are used in Section 4.1 to compare the standard code with the other NCBI translation tables.

	C			G			T			A		
C	CCC	Pro 115	62 53	CGC	Arg 174	94 80	CTC	Leu 131	72 59	CAC	His 155	82 73
	CCG	Pro 115	62 53	CGG	Arg 174	94 80	CTG	Leu 131	72 59	CAG	Gln 146	78 68
	CCT	Pro 115	62 53	CGT	Arg 174	94 80	CTT	Leu 131	72 59	CAT	His 155	82 73
	CCA	Pro 115	62 53	CGA	Arg 174	94 80	CTA	Leu 131	72 59	CAA	Gln 146	78 68
G	GCC	Ala 89	48 41	GGC	Gly 75	40 35	GTC	Val 117	64 53	GAC	Asp 133	70 63
	GCG	Ala 89	48 41	GGG	Gly 75	40 35	GTG	Val 117	64 53	GAG	Glu 147	78 69
	GCT	Ala 89	48 41	GGT	Gly 75	40 35	GTT	Val 117	64 53	GAT	Asp 133	70 63
	GCA	Ala 89	48 41	GGA	Gly 75	40 35	GTA	Val 117	64 53	GAA	Glu 147	78 69
T	TCC	Ser 105	56 49	TGC	Cys 121	64 57	TTC	Phe 165	88 77	TAC	Tyr 181	96 85
	TCG	Ser 105	56 49	TGG	Trp 204	108 96	TTG	Leu 131	72 59	TAG	STOP 0	0 0
	TCT	Ser 105	56 49	TGT	Cys 121	64 57	TTT	Phe 165	88 77	TAT	Tyr 181	96 85
	TCA	Ser 105	56 49	TGA	STOP 0	0 0	TTA	Leu 131	72 59	TAA	STOP 0	0 0
A	ACC	Thr 119	64 55	AGC	Ser 105	56 49	ATC	Ile 131	72 59	AAC	Asn 132	70 62
	ACG	Thr 119	64 55	AGG	Arg 174	94 80	ATG	Met 149	80 69	AAG	Lys 146	80 66
	ACT	Thr 119	64 55	AGT	Ser 105	56 49	ATT	Ile 131	72 59	AAT	Asn 132	70 62
	ACA	Thr 119	64 55	AGA	Arg 174	94 80	ATA	Ile 131	72 59	AAA	Lys 146	80 66

Figure 1: Standard genetic code with nucleon parameters of the encoded amino acids, displayed under the naive reading (Key 0), in which stop codons contribute zero and ATG is counted as methionine. In each cell, the codon appears on the left; the symbol of the encoded product (the three-letter amino acid symbol or STOP for the three stop codons) appears at the top center; the total nucleon count T appears below the product symbol; the proton count P appears at the top right; and the neutron count N appears at the bottom right. Codons of Octet I (the eight fully four-fold degenerate families) are shaded purple, and codons of Octet II are shaded yellow. Generated by the accompanying `visualization.html`.

2.3 Codon-group construction

In addition to the Rumer partition introduced in Section 2.2, we use an additional overlapping grouping system based on the chemical identity of the nucleotide at the third codon position.

Each of the four nucleotides belongs to one of two classes under each of three distinct binary chemical classifications:

- base type: *purine* $\{\mathrm{A}, \mathrm{G}\}$ (R) or *pyrimidine* $\{\mathrm{C}, \mathrm{T}\}$ (Y);
- number of hydrogen bonds in the canonical base pair: *strong* $\{\mathrm{C}, \mathrm{G}\}$ (S; three bonds) or *weak* $\{\mathrm{A}, \mathrm{T}\}$ (W; two bonds);
- functional class: *amino* $\{\mathrm{A}, \mathrm{C}\}$ (M) or *keto* $\{\mathrm{G}, \mathrm{T}\}$ (K).

These axes were not selected in response to the subsequent nucleon calculations: they are three conventional pairwise dichotomies of the nucleotides and are represented explicitly in the IUPAC–IUB ambiguity nomenclature by the symbol pairs R/Y, S/W, and M/K. Together they exhaust the three possible partitions of four nucleotides into two unordered pairs, thereby yielding six two-element subsets $X \subset \{\mathrm{A}, \mathrm{T}, \mathrm{G}, \mathrm{C}\}$.

For each of these six two-element subsets, and for each singleton $X \in \{\{\mathrm{A}\}, \{\mathrm{C}\}, \{\mathrm{G}\}, \{\mathrm{T}\}\}$, we define the codon group

$$G_X = \{(c_1, c_2, c_3) \in \mathcal{C}_{64} : c_3 \in X\}.$$

Thus, G_X consists of all codons whose third-position nucleotide belongs to X. The six two-nucleotide groups overlap: every codon belongs to exactly three of them, one for each chemical classification. At the same time, every codon belongs to exactly one of the four one-nucleotide groups.

This grouping system is applied at three levels: to all 64 codons, to the 32 codons of Octet I, and to the 32 codons of Octet II. If $|X| = 2$, then G_X contains 32 codons, while each of the intersections $G_X \cap$ Octet I and $G_X \cap$ Octet II contains 16 codons. If $|X| = 1$, the corresponding sizes are 16 and 8 codons. Group size alone does not identify a particular group; formulas therefore specify the set X and, where necessary, the octet explicitly.

The family of 33 codon groups used in the present work comprises the full code as one group of 64 codons; the two octets as groups of 32 codons each; ten groups G_X at the full-code level, corresponding to the six two-nucleotide and four one-nucleotide sets X; and their twenty intersections with Octet I and Octet II. Thus, the total number of analyzed groups is

$$1 + 2 + 10 + 20 = 33.$$

For example, $G_{\{\mathrm{G},\mathrm{T}\}} \cap$ Octet I is the 16-codon group Keto $\cap$ Octet I, whereas $G_{\{\mathrm{A}\}} \cap$ Octet II is the 8-codon group of Octet II with adenine at the third position. All 33 groups, together with their sizes, codon compositions, and Rumer-octet membership, are listed in the dataset `codon_groups.csv`.

In the group registry, the full set $\mathcal{C}_{64}$ corresponds to the row with id 1. In this article, that group is denoted All, whereas $\mathcal{C}_{64}$ denotes the codon set itself.

2.4 Pyrimidine synonymy and twin-group identities

Within every codon family of the standard genetic code, the codons NNC and NNT with identical first two positions encode the same amino acid. Therefore, exchanging $\mathrm{C} \leftrightarrow \mathrm{T}$ at the third position does not change the multiset of encoded amino acids. This exchange maps Keto $\{\mathrm{G}, \mathrm{T}\}$ to Strong $\{\mathrm{C}, \mathrm{G}\}$, and Amino $\{\mathrm{A}, \mathrm{C}\}$ to Weak $\{\mathrm{A}, \mathrm{T}\}$.

The two members of each pair contain the same service codons: Keto and Strong contain ATG and TAG, whereas Amino and Weak contain TAA and TGA. Therefore, under any assignment K that associates a pair of values (P, N) with each service codon (Section 2.5) and for $Q \in \{T, P, N, \Delta\}$, the identities

$$Q_K(\text{Keto}) = Q_K(\text{Strong}), \qquad Q_K(\text{Amino}) = Q_K(\text{Weak})$$

hold. Intersecting both groups in either pair with the same octet preserves these identities. The equalities are independent of parametrization, although the full-group values themselves may depend on K. We refer to the Keto/Strong and Amino/Weak pairs, and to the corresponding pairs formed by intersection with the same octet, as *twin groups*. In the dataset `codon_groups.csv`, this relationship is recorded in the `Twin_Group_ID` column.

Purine and Pyrimidine are not twin groups: exchanging C $\leftrightarrow$ T preserves each of these groups rather than mapping one to the other. The same pyrimidine synonymy directly gives equality of all four nucleon quantities for the one-nucleotide groups {C} and {T} and for their intersections with either octet. These groups and their intersections with the octets contain no service codons, so both the values themselves and the equalities between them are independent of parametrization. Because this is a pair of one-nucleotide groups rather than a pair of two-nucleotide branches of the chemical axes, the term *twin groups* and the `Twin_Group_ID` column are not used for it.

2.5 Parametrization of service codons

The nucleon quantities of a codon group are obtained by summing the values of the amino acids encoded by its constituent codons in the standard genetic code. For codons counted only according to their amino-acid assignments, this summation is unambiguous. A separate convention is needed for the three stop codons, which have no amino-acid product, and for ATG in its initiator role; ATG also encodes methionine (Section 2.2).

The complete codon set $\mathcal{C}_{64}$, the four-codon service set $\mathcal{S}$, and the remaining 60-codon pool $\mathcal{C}_{\mathrm{SE}}$ are defined in Section 2.2. We refer to $\mathcal{C}_{\mathrm{SE}}$ below as the *service-excluded pool*; its technical identifier in the code and file names is `pool60`. Every codon in $\mathcal{C}_{\mathrm{SE}}$ is unambiguously assigned to a canonical amino acid. For $c \in \mathcal{C}_{\mathrm{SE}}$ and $Q \in \{T, P, N, \Delta\}$, $Q(c)$ denotes the corresponding quantity of the free neutral form of the amino acid encoded by c, as defined in Section 2.1.

For any codon set $A \subseteq \mathcal{C}_{64}$ containing no service codons, define

$$Q(A) := \sum_{c \in A} Q(c).$$

For a codon group G, its sum over the service-excluded pool is defined by

$$Q_{\mathrm{SE}}(G) := \sum_{c \in G \cap \mathcal{C}_{\mathrm{SE}}} Q(c).$$

For a service codon c, the notation $Q(c)$ without a key subscript is not used because the formal contribution of such a codon depends on the selected key. The subscript SE indicates a restriction of the summation domain, not a parametrization key. Neither the sum $Q_{\mathrm{SE}}(G)$ nor the number of codons included in it, $|G \cap \mathcal{C}_{\mathrm{SE}}|$, depends on the assignments to the service codons.

If the group G itself contains no service codons, the restriction removes nothing and

$$Q(G) = Q_{\mathrm{SE}}(G).$$

To analyze the complete 64-position table, we introduce a *parametric key* K: a fixed assignment of a pair of numbers $\big(P_K(c), N_K(c)\big)$ to every service codon c. Under such a key, the group sums are defined by

$$\big(P_K(G), N_K(G)\big) := \big(P_{\mathrm{SE}}(G), N_{\mathrm{SE}}(G)\big) + \sum_{c \in G \cap \mathcal{S}} \big(P_K(c), N_K(c)\big),$$

and the remaining two quantities are derived rather than assigned independently:

$$T_K(G) = P_K(G) + N_K(G), \qquad \Delta_K(G) = P_K(G) - N_K(G).$$

The same two relations define $T_K(c)$ and $\Delta_K(c)$ for each service codon c. For $Q \in \{T, P, N, \Delta\}$, $Q_0(G)$ and $Q_1(G)$ below denote the full group sums, including service-codon contributions,

under the Key 0 and Key 1 assignments, respectively. The subscripts 0 and 1 label these two assignments; they are not numerical values of a scalar parameter. Thus $Q_{\mathrm{SE}}(G)$ is independent of the key, whereas $Q_0(G)$ and $Q_1(G)$ can differ only through the contributions of the service codons contained in group G. If group G contains no service codons, then $Q_K(G) = Q(G) = Q_{\mathrm{SE}}(G)$ for every K. The subscript SE applies only to codon groups; the subscripts K, 0, and 1 apply to full group sums and to the formal contributions of individual service codons. The two keys considered in this study are recorded in `key_parameters.csv`.

The naive reading (Key 0). Each of the three stop codons is assigned the zero contribution $(P, N) = (0, 0)$ because a termination signal has no amino acid product. The ATG codon is treated as an ordinary methionine codon and receives the values of the free methionine molecule, $(P, N) = (80, 69)$. This simple convention corresponds to the ordinary reading of the amino-acid-coding part of the table, in which the stop signals are assigned a zero contribution, but it remains a parametrization. In particular, Key 0 is not identical to the service-excluded pool: Key 0 retains all 64 positions and counts ATG as methionine, whereas in the pool the summation runs over the remaining 60 codons only.

Symmetrization (Key 1). Each of the three stop codons is assigned $(P, N) = (37, 37)$, while ATG, treated as a formal initiation marker, is assigned $(P, N) = (1, 0)$. These values are not interpreted as the nucleon composition of physical translation products. They are derived in Sections 3.4–3.4.3 under the assumption that the three stop codons share a common parameter pair and under the conditions of balance and divisibility by 37 specified there. Under these assumptions, Key 1 is the unique minimal non-negative integer solution of the resulting system. Minimality here means the smallest total contribution of the service codons—the sum of the contributions of the three stop codons and ATG—determined separately for protons and for neutrons. For the standard code the balance fixes the difference between the stop and start parameters, so the total contribution increases with the start parameter and is minimal at its smallest admissible value. Here Key 1 is introduced as the result of that formal construction, not as a biochemical property of termination signals or the initiator.

For codon groups containing no service codons, the values under Key 0 and Key 1 coincide with the direct amino-acid sums. If a group contains service codons, the Key 1 assignments have only a formal arithmetic meaning: the corresponding sums do not represent the nucleon composition of a physical ensemble of molecules. The notation P, N, T, and Δ is retained for consistency with the underlying amino acid contributions, but no biochemical interpretation is claimed for the formal Key 1 contributions.

The dataset `key_parameters.csv` additionally contains eight reference assignments, `Key 2`–`Key 9`, considered during the exploratory stage. They are not used to obtain the results of this study and do not have the status of the derived Key 1. The following analysis therefore distinguishes three levels explicitly: the service-excluded pool, the naive reading (Key 0), and the symmetrization (Key 1).

2.6 Analytical criteria and classification of equalities

For each of the 33 codon groups defined in Section 2.3, we examine P, N, T, and Δ at three analytical levels: the service-excluded pool, Key 0, and Key 1. Three reporting procedures are described below: tests of divisibility by 37 and decimal roundness, and a search for exact equalities. Divisibility by 37 is the primary preselected arithmetic criterion. Exact powers, reduced ratios, and matches to amino acid differences are considered separately as additional descriptive observations. Statistical significance is assessed only where a corresponding null model is specified.

Criterion 1: divisibility by 37. For every group quantity P, N, T and Δ, we compute the quotient and remainder upon division by 37. A zero remainder denotes exact divisibility. This criterion continues the line of investigation opened by Shcherbak and Makukov [8] and developed by Panov and Filatov [5]. When a value is divisible by 37, its displayed factorization also exposes any further factors of the quotient; these are not assigned separate evidential weight without an additional analysis. Nonzero residues are reported descriptively only when they enter exact relations among predefined groups or form recurring patterns of one-unit deviations from the nearest multiple of 37.

Complete expressions of the form $q \cdot 37 + r$ for all groups and all four quantities are provided in `pool60_divisibility-37.csv`, `key0_divisibility-37.csv` and `key1_divisibility-37.csv`.

Criterion 2: roundness in the decimal system. We call a positive integer quantity round in the decimal system if it is divisible by 10^k for at least one integer $k \geq 1$; its degree of roundness is the largest such k. This criterion is used descriptively because the data contain exact multiples of 100 and 1000, notably among the parametrization-independent Octet I sums. Although roundness is tied to decimal notation, the divisibility being tested is a property of the integer itself: $10^k \mid x$ is equivalent to simultaneous divisibility by 2^k and 5^k. The systematic control across prime moduli is presented in Section 4.2.

Criterion 3: exact equalities. At each analytical level, we compare the P, N, T, and Δ values of all 33 groups and record every exact match. Trivial identities such as $P(G) = P(G)$ are not included in the results. Each equality is assigned to one of three classes:

- *different-size equalities* (`DIFF_PARAM` in the `Match_Type` column): the same numerical value is obtained by summing over different numbers of codons; either the same or different nucleon quantities may be compared;
- *same-size cross-quantity equalities* (`CROSS_PARAM`): the groups contain the same number of codons, but different quantities are compared, for example $P(G_1) = N(G_2)$;
- *same-size same-quantity equalities* (`SAME_PARAM`): the groups contain the same number of codons and the same quantity is compared, for example $P(G_1) = P(G_2)$.

For classification, group size is the number of codons actually included in the sum. Under Key 0 and Key 1, this is the full size of the group in the 64-codon table. In the service-excluded pool, the service codons are removed, so groups containing them are smaller.

All identified equalities are listed in `pool60_equalities.csv`, `key0_equalities.csv` and `key1_equalities.csv`. Twin-group identities from Section 2.4 are also recorded, but are treated as consequences of identical amino acid composition rather than as new coincidences.

Additional exploratory observations. Beyond the primary procedures, we descriptively note:

- exact powers of small integers, for example $4096 = 2^{12}$;
- reduced ratios within a group, including P/T, N/T, Δ/T, P/N, Δ/P and Δ/N, tabulated in `pool60_ratios.csv`, `key0_ratios.csv` and `key1_ratios.csv`;
- matches between group-level quantities and the absolute differences $|\delta P|$, $|\delta N|$ or $|\delta T|$ of individual amino acid pairs, such as $|\delta N(\mathrm{Trp}, \mathrm{Ile})| = 37$.

These observations describe the arithmetic and may provide a basis for testable hypotheses.

Finally, the four quantities are not independent:

$$T = P + N, \qquad \Delta = P - N.$$

Equalities, divisibility relations or ratios obtained from one another through these identities are therefore not counted as separate evidence. The same applies to copies propagated through nested groups or twin-group identities. Section 3 distinguishes (i) facts computed directly for the service-excluded pool without assigning values to the service codons; (ii) identities whose invariance to the key is established algebraically; and (iii) results specific to Key 0 or Key 1.

2.7 Computed datasets and reproducibility

The principal deterministic results of this study are produced by `reproduce.py`. The script reads five input datasets and writes 18 computed output datasets. A separate deterministic script, `controls.py`, writes four control datasets. The main statistical calculation is performed by `mc_significance.py`, and the additional checks by `mc_sensitivity.py`. They write `mc_significance.csv` and `mc_sensitivity.csv`, respectively. The results are independently verified by `verify_mc.py` and `verify_mc_sensitivity.py`. By default, all data paths are resolved relative to the directory of the corresponding script.

Input datasets. Five input files define the numerical and structural basis of the analysis:

- `amino_acids_nucleons.csv` contains P, N, T and Δ for the twenty canonical amino acids (Section 2.1);
- `genetic_code_codons.csv` specifies the standard codon-to-translation-product mapping (Section 2.2);
- `ncbi_genetic_code_registry.csv` records a snapshot of the 27 NCBI translation tables, including the standard table (Section 2.2);
- `codon_groups.csv` defines the system of 33 overlapping codon groups (Section 2.3);
- `key_parameters.csv` contains the service-codon assignments of the parametric keys (Section 2.5).

Per-key computed datasets. For each of the two principal keys, Key 0 and Key 1, `reproduce.py` writes four files. The prefixes `key0_` and `key1_` identify the active key:

- `*_nucleon-data.csv` contains the four reported group quantities P, N, T and Δ for all 33 groups; T and Δ are derived from P and N;
- `*_divisibility-37.csv` expresses every quantity exactly as $q \cdot 37 + r$;
- `*_equalities.csv` exhaustively enumerates exact numerical equalities and classifies them in the `Match_Type` column as `DIFF_PARAM`, `CROSS_PARAM` or `SAME_PARAM`, following Section 2.6;
- `*_ratios.csv` contains the six ratios P/T, N/T, Δ/T, P/N, Δ/P and Δ/N both as exact reduced fractions and as rounded decimal approximations.

The two keys therefore produce eight output files.

Service-excluded-pool datasets. For the service-excluded pool, the script writes four analogous files: `pool60_nucleon-data.csv`, `pool60_divisibility-37.csv`, `pool60_equalities.csv` and `pool60_ratios.csv`. Before summation, the service codons are removed from every group G, so that the number of codons included is $|G \cap \mathcal{C}_{\mathrm{SE}}|$. No parametric key is applied.

Amino acid differences. Three files written once, `amino_acids_proton_differences.csv`, `amino_acids_neutron_differences.csv` and `amino_acids_nucleon_differences.csv`, enumerate the absolute differences $|\delta P|$, $|\delta N|$ and $|\delta T|$, respectively, for all $\binom{20}{2} = 190$ unordered pairs of canonical amino acids.

Table-dependent NCBI pools. The remaining three outputs of the core script compare the standard code with the other tables in `ncbi_genetic_code_registry.csv`. Because the sets of stop codons differ between tables, we do not assume in advance that every analyzed pool contains 60 codons. Two models are used. Model 1 excludes ATG and every codon listed in the `Stop_Codons` field. Model 2 excludes ATG and only the permanent stop codons marked `*` in the `Amino_Acids` field; context-dependent stops are included in the analysis with the amino acid assignments recorded for them. The models coincide for tables without context-dependent stops. For the standard table, both models give the same 60-codon service-excluded pool. The actual number of codons in the analyzed pool is recorded for both models (columns `Pure_Codons_Count` and `Pure_Codons_Count_Alt`).

`deficit_models_analysis.csv` contains one row for each of the 27 tables. The full proton and neutron sums follow the registered amino acid assignments, with positions marked `*` contributing zero. For both models, the file additionally records sums over the corresponding analyzed pool, the proton-sum residue modulo 37, and three neutron deficits relative to:

1. the constant 3700 obtained for standard-code Octet I;
2. structural T(Octet I), retaining the eight families that occupy Octet I positions in the standard table;
3. functional T(Octet I), selecting in table t those four-codon families that have one amino acid assignment and no permanent stop codon.

`keto_amino_balance_models.csv` uses the same two models and, for every table, sums P and N separately over the intersections of the analyzed pool with the third-position Keto and Amino groups. Both sums and the directed Keto-minus-Amino difference are recorded.

`service_codon_balance_comparison.csv` contains 54 rows: one for each table and each model. For protons and neutrons separately, non-negative integer assignments are determined by minimizing the total contribution of the service codons. All stop codons in the chosen model receive the same value, while ATG receives a separate value. Two procedures are compared: Keto/Amino balance alone and the same balance with the additional requirement that the complete-table sum be divisible by 37. The file records the assignments, added contributions, resulting sums, and the difference between the minimal contributions under the two procedures. It distinguishes a unique assignment, a determined total contribution without a unique assignment, and the absence of a solution. The application of these procedures to the standard code is presented in Section 3.4.

Reproducibility of the core pipeline. `reproduce.py` uses only the Python standard library and requires neither external packages nor a virtual environment. The computation is deterministic: outputs are written in UTF-8 with LF line endings and a fixed row order. The canonical repository files are regenerated from the five inputs without manual editing. All group sums and residues use integer arithmetic, while ratios are first reduced exactly by their greatest common divisor. The decimal columns in `*_ratios.csv` are rounded representations of exact fractions and are not used to obtain any result. Every deterministic numerical claim can be traced to a specific cell or to a documented exact combination of cells in the output files.

Neither `reproduce.py` nor `controls.py` uses randomization. Randomization is used in the main Monte Carlo calculation (`mc_significance.py`), the additional checks (`mc_sensitivity.py`), and their independent implementations (`verify_mc.py` and `verify_mc_`

`sensitivity.py`). Random seeds and numbers of trials are specified in all cases. Tail-probability estimates and intervals use floating-point values; tail membership is determined without rounding the statistics. The statistical procedures are described in Section 4.3.

Supplementary visualizer. `visualization.html` provides an interactive view of the translation tables and codon groups under the service-excluded pool (POOL60), Key 0 and Key 1, together with additional exploratory views. It is not part of the core reproducible pipeline and is not the source of published numerical results: the deterministic numerical values reported in the main text are checked against the corresponding CSV files. An online version is available at https://ruslankhafizov.github.io/genetic-code-arithmetic/.

Supplementary control datasets. `controls.py` reads the same five input files, uses only the Python standard library and writes four deterministic datasets:

- `position_axis_analysis.csv` separately applies the three binary chemical classifications Keto/Amino, Strong/Weak and Purine/Pyrimidine to the first, second and third codon positions at three analytical levels: the service-excluded pool, Key 0 and Key 1. For each side it records the four nucleon quantities and their residues modulo 37;
- `prime_divisibility_scan.csv` scans every prime up to 97 over three value sets: group quantities identical under Key 0 and Key 1, all Key 0 quantities, and all Key 1 quantities. For each prime it records the numbers of divisible instances, distinct nonzero values and distinct odd cores, together with a reference value n/p for n distinct values. This reference describes residue uniformity and is not a statistical null model or a p-value;
- `position_asymmetry_ncbi.csv` applies the Keto/Amino classification to each of the three positions of all 27 NCBI tables under Model 1 and Model 2. For T, P, N and Δ, it records directed Keto-minus-Amino differences, their residues modulo 37 and position-level summary flags;
- `representation_sensitivity.csv` recomputes 13 explicitly listed absolute and difference quantities under three representations of the encoded amino acid. Relative to the neutral free molecule, the proton and neutron offsets are $(0, 0)$ for the free molecule itself, $(-10, -8)$ for the peptide residue, and $(0, 0)$ for the zwitterion. For every quantity, the file records the result under each representation and its agreement with the free-molecule baseline.

These four files contain the results of three controls—codon position, choice of modulus, and molecular representation—and of the cross-table comparison of Keto/Amino differences. The interpretation is given in Section 4.2.

Data and code availability. The accompanying repository contains the manuscript sources, five input datasets, 18 core computed datasets, four control datasets, the two Monte Carlo summary tables `mc_significance.csv` and `mc_sensitivity.csv`, six Python scripts and the interactive visualizer. Complete run commands and descriptions of the dataset columns are provided in the repository README: https://github.com/RuslanKhafizov/genetic-code-arithmetic. Versions of the materials are archived at Zenodo under the concept DOI 10.5281/zenodo.20360037.

3 Results

The arithmetic properties are considered here at three analytical levels. The first level (Section 3.1) comprises facts established without assigning values to the service codons.

The second level (Section 3.2) is the naive reading (Key 0), in which the three stop codons are assigned a contribution of zero and the initiator ATG is counted as methionine.

Section 3.3 reduces the asymmetry of the Keto and Amino branches to the nucleon differences of the Trp–Ile pair and examines the parity constraint on the proton quantities.

At the third level, Section 3.4 derives Key 1, the unique minimal non-negative integer assignment to the service codons with equal stop-codon contributions, determined by the joint conditions of nucleon-pool balance along the Keto/Amino axis and divisibility of the global pools by 37; equivalently, this balance augments the C/T symmetry imposed by the code with G/A symmetry in proton and neutron sums grouped by the nucleotide in the third position.

Section 3.5 uses this formal assignment to extend the arithmetic analysis to the complete 64-codon table and distinguishes between identities that follow directly from the derivation conditions and additional regularities not used to select Key 1.

Section 3.6 draws together the findings of all three levels.

The standard code remains the principal object of analysis throughout. Alternative NCBI translation tables are examined later for comparison and to assess the specificity of the results.

3.1 Parametrization-independent results

The parametrization-independent results for group-level nucleon quantities in this subsection are obtained under one of two conditions: either the summation domain is restricted to the service-excluded pool, or the group under consideration itself contains no service codons.

These results form the basis of the arithmetic structure that is traced further as the service codons are restored. The neutron sum of the service-excluded pool is divisible by 37, whereas the proton sum falls one unit short of the nearest multiple. Through the difference Δ this deviation is traced to the codons with G at the third position within Octet II — the group that contains the initiator ATG in the full table (Section 3.1.1). The nucleon sum of Octet I is divisible by 37, and all four of its quantities are divisible by 100 (Section 3.1.2). For the pyrimidine-restricted groups of the full code the difference Δ lies on the lattice of step 37, whereas the deviations arising on intersection with the octets cancel within the group (Section 3.1.3).

3.1.1 The service-excluded pool

The Keto/Amino and Strong/Weak partitions of $\mathcal{C}_{\mathrm{SE}}$ give pairwise equal neutron and proton sums, owing to pyrimidine synonymy (Section 2.4):

$$\begin{aligned} N_{\mathrm{SE}}(\mathrm{Keto}) = N_{\mathrm{SE}}(\mathrm{Strong}) &= 1813 = 49 \cdot 37, \\ N_{\mathrm{SE}}(\mathrm{Amino}) = N_{\mathrm{SE}}(\mathrm{Weak}) &= 1776 = 48 \cdot 37, \\ P_{\mathrm{SE}}(\mathrm{Keto}) = P_{\mathrm{SE}}(\mathrm{Strong}) &= 2108 = 57 \cdot 37 - 1, \\ P_{\mathrm{SE}}(\mathrm{Amino}) = P_{\mathrm{SE}}(\mathrm{Weak}) &= 2072 = 56 \cdot 37. \end{aligned}$$

Both neutron sums lie on the 37-lattice separately. The proton sum of the Amino half lands on it exactly, whereas the Keto half falls one unit short of the nearest multiple. Adding the halves gives, for the pool as a whole,

$$N(\mathcal{C}_{\mathrm{SE}}) = 3589 = 97 \cdot 37, \qquad P(\mathcal{C}_{\mathrm{SE}}) = 4180 = 113 \cdot 37 - 1, \qquad \Delta(\mathcal{C}_{\mathrm{SE}}) = 591 = 16 \cdot 37 - 1.$$

Analysis of Δ localizes the one-unit deviation of the proton sum of the pool from the nearest multiple of 37 in the group $\{\mathrm{G}\} \cap$ Octet II. On the Keto/Amino and Strong/Weak axes the neutron sums are multiples of 37, so $\Delta = P - N$ has the same residue as the proton sum. On the Purine/Pyrimidine axis neither P nor N lies on the lattice, but their residues coincide for Pyrimidine and differ by one for Purine:

$$\begin{aligned} \Delta_{\mathrm{SE}}(\mathrm{Keto}) = \Delta_{\mathrm{SE}}(\mathrm{Strong}) = \Delta_{\mathrm{SE}}(\mathrm{Purine}) &= 295 = 8 \cdot 37 - 1, \\ \Delta_{\mathrm{SE}}(\mathrm{Amino}) = \Delta_{\mathrm{SE}}(\mathrm{Weak}) = \Delta(\mathrm{Pyrimidine}) &= 296 = 8 \cdot 37. \end{aligned}$$

Keto, Strong, Amino, and Weak each contain 30 codons in the pool, Purine contains 28, and Pyrimidine contains 32. The same two values of Δ, 295 and 296, arise in all three partitions — both for parts of equal size 30/30 and for the unequal split 28/32.

At the level of the one-nucleotide groups none of the sums P and N is a multiple of 37, and the one-unit deviation in Δ is carried by one group — the branch {G}, common to Keto, Strong, and Purine:

$$\Delta_{\mathrm{SE}}(\{\mathrm{G}\}) = 147 = 4 \cdot 37 - 1, \qquad \Delta_{\mathrm{SE}}(\{\mathrm{A}\}) = \Delta(\{\mathrm{C}\}) = \Delta(\{\mathrm{T}\}) = 148 = 4 \cdot 37.$$

Keto, Strong, and Purine are the unions of {G} with {T}, {C}, and {A} respectively. The one-unit deviation localized in $\Delta_{\mathrm{SE}}(\{\mathrm{G}\})$ can therefore be traced across all three axes.

Within Octet I all four one-nucleotide groups have $\Delta = 75$, so the difference falls entirely in Octet II:

$$\Delta_{\mathrm{SE}}(\{\mathrm{G}\} \cap \text{Octet II}) = 72, \qquad \Delta_{\mathrm{SE}}(\{\mathrm{A}\} \cap \text{Octet II}) = \Delta(\{\mathrm{C}\} \cap \text{Octet II}) = \Delta(\{\mathrm{T}\} \cap \text{Octet II}) = 73.$$

In the pool the groups $\{\mathrm{G}\} \cap$ Octet II and $\{\mathrm{A}\} \cap$ Octet II contain six codons each, whereas $\{\mathrm{C}\} \cap$ Octet II and $\{\mathrm{T}\} \cap$ Octet II contain eight. The observed distribution of the values 72 and 73 therefore cannot be attributed to group size.

On passing to the complete 64-codon table, these two six-codon branches are supplemented by different pairs of service codons: $\{\mathrm{A}\} \cap$ Octet II by the stop codons TAA and TGA, and $\{\mathrm{G}\} \cap$ Octet II by the stop codon TAG and the initiator ATG. As will be shown in the derivation of Key 1 (Section 3.4), the formal Δ-contribution under that assignment equals 1 for the initiator ATG and 0 for each stop codon. Therefore, under Key 1,

$$\Delta_1(\{\mathrm{G}\} \cap \text{Octet II}) = 72 + 1 + 0 = 73, \qquad \Delta_1(\{\mathrm{A}\} \cap \text{Octet II}) = 73 + 0 + 0 = 73.$$

The missing unit is thus returned through the initiator ATG, aligning the two branches. Since ATG belongs to the branch {G} and to Keto, Strong, and Purine, the same single contribution simultaneously compensates for the deviation in all of these groups and in the value of Δ for the complete table under Key 1.

The full hierarchy of groups for the service-excluded pool, together with the corresponding nucleon quantities, is shown in Figure 2.

3.1.2 Arithmetic of Octet I

Octet I (id 12) contains no service codons, so its nucleon quantities are independent of any parametrization:

$$T(\text{Octet I}) = 3700 = 100 \cdot 37, \quad P(\text{Octet I}) = 2000 = 2 \cdot 1000,$$
$$N(\text{Octet I}) = 1700 = 17 \cdot 100, \quad \Delta(\text{Octet I}) = 300 = 3 \cdot 100.$$

All four quantities are multiples of 100, while the proton sum P(Octet I) is also a multiple of 1000. In addition, the nucleon sum T(Octet I) is divisible by 37.

Octet I consists of eight fully four-fold degenerate codon families. Each of its six 16-codon groups (ids 13–18) contains two codons from every family, so each of its nucleon quantities equals half the corresponding quantity of the whole octet:

$$T = 1850 = 50 \cdot 37, \qquad P = 1000 = 10^3, \qquad N = 850, \qquad \Delta = 150.$$

Each of its four 8-codon groups (ids 19–22) contains one codon from every family, so its quantities equal one quarter of the whole-octet values:

$$T = 925 = 25 \cdot 37, \qquad P = 500 = 5 \cdot 100, \qquad N = 425, \qquad \Delta = 75 = 2 \cdot 37 + 1.$$



Figure 2: Structure of the service-codon-excluded 60-codon pool $\mathcal{C}_{\mathrm{SE}}$ (`pool60`). For each group, all four quantities are computed using only the non-service codons it contains; the three stop codons and the initiator ATG are excluded, so every quantity shown is independent of parametrization. The quantities shown are the total nucleon sum T, proton sum P, neutron sum N, and hydrogen atom count $\Delta = P - N$. Cell layout: the group name appears in the upper-left corner, the number of codons remaining in the group in the upper-right corner, and the identifier from `codon_groups.csv` in the lower-right corner; twin pairs of groups with identical values are collapsed into a single cell, with the second name in the lower-left corner and both identifiers listed, separated by a comma, in the lower-right corner. Green, blue, and purple highlights indicate divisibility by 37, $111 = 3 \cdot 37$, and $999 = 27 \cdot 37$, respectively; the largest applicable modulus is marked. Figures 3 and 4 use the same grid and color code, but represent the complete 64-codon table. All three figures were generated by the accompanying `visualization.html`.

Thus, at all three levels of the Octet I hierarchy, exact scaling preserves divisibility of the nucleon sum T by 37 and a single rational profile with a common denominator of 37:

$$\frac{P}{T} = \frac{20}{37}, \qquad \frac{N}{T} = \frac{17}{37}, \qquad \frac{\Delta}{T} = \frac{3}{37}.$$

The proton and neutron sums of the whole octet have residues $+2$ and -2 modulo 37, which cancel in $T = P + N$. The neutron residue -2 of Octet I is, in turn, compensated by the residue $+2$ of the Octet II neutron sum in the service-excluded pool. Adding the neutron sums of the two octets gives the pool neutron sum $N(\mathcal{C}_{\mathrm{SE}})$ obtained in Section 3.1.1. Both cancellations are represented by the following equalities:

$$\begin{aligned} P(\text{Octet I}) &= 2000 = 54 \cdot 37 + 2, \\ N(\text{Octet I}) &= 1700 = 46 \cdot 37 - 2, \\ N_{\mathrm{SE}}(\text{Octet II}) &= 1889 = 51 \cdot 37 + 2, \\ N(\mathcal{C}_{\mathrm{SE}}) &= N(\text{Octet I}) + N_{\mathrm{SE}}(\text{Octet II}) \\ &= 1700 + 1889 = 3589 = 97 \cdot 37. \end{aligned}$$

At the 16-codon level, exact halving gives residues $+1$ for P and -1 for N modulo 37. Of the four 8-codon groups, the two pyrimidine groups— $\{\mathrm{C}\} \cap$ Octet I (id 19) and $\{\mathrm{T}\} \cap$ Octet I (id 21)—form the Pyrimidine group within Octet I (id 18), which enters the pyrimidine cascade in Section 3.1.3.

3.1.3 Arithmetic of pyrimidine-restricted groups

By pyrimidine synonymy (Section 2.4), the {C} and {T} groups have identical nucleon quantities at the full-code level and after intersection with either octet. Together they form the Pyrimidine group (id 7). These groups and their octet intersections contain no service codons, so their values are independent of parametrization. Unlike the Octet I hierarchy, the full Pyrimidine group spans both octets and includes partially degenerate codon families. Therefore, not all values at its lower levels are obtained by a single exact scaling; they are examined separately below.

For the full Pyrimidine group, the nucleon quantities are

$$\begin{aligned} T(\text{Pyrimidine}) &= 4096 = 2^{12}, & P(\text{Pyrimidine}) &= 2196, \\ N(\text{Pyrimidine}) &= 1900 = 19 \cdot 100, & \Delta(\text{Pyrimidine}) &= 296 = 8 \cdot 37. \end{aligned}$$

Here, Δ satisfies the primary preselected criterion of divisibility by 37, and $N = 1900$ satisfies the roundness criterion. The exact power of two $T = 2^{12}$ is an additional descriptive observation.

By the C/T identity, the two 16-codon groups {C} (id 8) and {T} (id 10) have identical nucleon quantities. Because they partition the Pyrimidine group, each value is half the corresponding full-group value:

$$\begin{aligned} T(\{\mathrm{C}\}) = T(\{\mathrm{T}\}) &= 2048 = 2^{11}, & P(\{\mathrm{C}\}) = P(\{\mathrm{T}\}) &= 1098, \\ N(\{\mathrm{C}\}) = N(\{\mathrm{T}\}) &= 950, & \Delta(\{\mathrm{C}\}) = \Delta(\{\mathrm{T}\}) &= 148 = 4 \cdot 37. \end{aligned}$$

The entire profile at this level is therefore obtained by exact scaling and does not form a separate set of independent regularities.

At the 8-codon level, all four Octet I groups, including $\{\mathrm{C}\} \cap$ Octet I (id 19) and $\{\mathrm{T}\} \cap$ Octet I (id 21), have $\Delta = 75$ (Section 3.1.2). The corresponding $\{\mathrm{C}\} \cap$ Octet II (id 30) and $\{\mathrm{T}\} \cap$ Octet II (id 32) groups have $\Delta = 73$. Their unions within the octets form the Pyrimidine $\cap$ Octet I (id 18)

and Pyrimidine $\cap$ Octet II (id 29) groups. These values form the following cascade:

$$\begin{aligned}
\Delta(\{\mathrm{C}\} \cap \text{Octet I}) = \Delta(\{\mathrm{T}\} \cap \text{Octet I}) &= 75 = 2 \cdot 37 + 1, \\
\Delta(\{\mathrm{C}\} \cap \text{Octet II}) = \Delta(\{\mathrm{T}\} \cap \text{Octet II}) &= 73 = 2 \cdot 37 - 1, \\
\Delta(\text{Pyrimidine} \cap \text{Octet I}) &= 2 \cdot 75 = 150 = 4 \cdot 37 + 2, \\
\Delta(\text{Pyrimidine} \cap \text{Octet II}) &= 2 \cdot 73 = 146 = 4 \cdot 37 - 2, \\
\Delta(\text{Pyrimidine}) &= 150 + 146 = 296 = 8 \cdot 37.
\end{aligned}$$

The one-unit deviations of opposite sign, $+1$ and -1, double to $+2$ and -2 when the groups within each octet are added, and cancel when the octets are added. Thus, the values 150, 146, and 296 belong to one additive scheme rather than constituting independent results.

The relation of the pyrimidine cascade to the pattern of one-unit deviations in the proton and neutron sums is considered under Key 0 (Section 3.2.3).

The pyrimidine cascade should also be distinguished from the one-unit deviation of $\Delta_{\mathrm{SE}}(\{\mathrm{G}\})$ from the lattice of step 37 (Section 3.1.1). The pyrimidine deviations cancel between the octets, whereas the deviation of the {G} group is not compensated when it is combined with another one-nucleotide group and propagates to the Keto, Strong, and Purine groups. Because the neutron sums of Keto and Strong in the service-excluded pool are multiples of 37, the identity $P = N + \Delta$ implies that their proton sums have the same one-unit deviation, -1.

3.2 Results under Key 0

Under the naive reading (Key 0), summation is performed over the complete 64-codon table: the stop-codon contribution is zero, whereas the initiator ATG is counted as methionine (Section 2.5). Thus, in passing from the service-excluded pool to the complete table, the only nonzero contribution comes from ATG:

$$\big(T(\text{Met}), P(\text{Met}), N(\text{Met}), \Delta(\text{Met})\big) = (149, 80, 69, 11).$$

The deviations from the nearest multiples of 37 for the service-excluded pool, methionine, and the complete 64-codon table are related as

$$(-1, -1, 0, -1) + (+1, +6, -5, +11) = (0, +5, -5, +10).$$

The ATG contribution brings the nucleon sum T exactly onto the 37-lattice, since $149 = 4 \cdot 37 + 1$ exactly compensates the one-unit deviation -1, but at the same time moves the neutron sum off the lattice, although it is divisible by 37 in the pool (Section 3.1.1).

Below, we show how this transformation propagates through the nested groups containing ATG. In the Amino and Weak groups, all the restored positions are occupied by stop codons, so the sums of the complete 32-codon groups coincide numerically with the sums over the 30 codons in the corresponding branches of the service-excluded pool. These sums already lie on the 37-lattice (Section 3.1.1), and the divisibility is therefore preserved in passing to the complete table. Under Key 0, restoring the service positions makes the groups on the two sides of the mirror pattern of one-unit deviations equal in size. The pattern appears at two levels: at the 16-codon level in the proton and neutron sums, and at the 8-codon level in the difference Δ. Within Octet II at the 8-codon level, this pattern, established earlier for the pyrimidine groups (Section 3.1.3), also extends to the {A} group: its value of Δ already coincided with theirs in the pool, while restoring the service positions makes all three groups equal in size. The sole exception is the {G} branch, which contains ATG.

The overall pattern of group quantities under Key 0 is shown in Fig. 3.



Figure 3: Nucleon quantities T, P, N, and Δ for all codon groups under Key 0. The grid, twin-group pair merging, and divisibility color code are the same as in Fig. 2, but the quantities are calculated for the complete 64-codon table. Generated by the accompanying `visualization.html`.

3.2.1 Divisibility of T by 37 at the upper levels of partitioning

The ATG contribution falls in Octet II. Under Key 0, its four 8-codon groups have the following profile:

$$\begin{aligned} T(\{\mathrm{C}\} \cap \text{Octet II}) = T(\{\mathrm{T}\} \cap \text{Octet II}) &= 1123 = 30 \cdot 37 + 13, \\ T_0(\{\mathrm{A}\} \cap \text{Octet II}) &= 875 = 24 \cdot 37 - 13, \\ T_0(\{\mathrm{G}\} \cap \text{Octet II}) &= 1097 = 30 \cdot 37 - 13. \end{aligned}$$

The pyrimidine branches deviate from the nearest multiple by $+13$, the purine branches by -13. In the service-excluded pool, the deviation of the $\{\mathrm{G}\}$ branch differs by one unit from that of the $\{\mathrm{A}\}$ branch:

$$T_{\mathrm{SE}}(\{\mathrm{G}\} \cap \text{Octet II}) = 948 = 26 \cdot 37 - 14.$$

The ATG contribution changes the deviation of the $\{\mathrm{G}\}$ branch from -14 to -13, so that both purine branches have the same deviation.

Among the groups of Octet II, the nucleon sum is divisible by 37 exactly for those that contain equal numbers of purine and pyrimidine branches:

$$\begin{aligned} T_0(\text{Keto} \cap \text{Octet II}) = T_0(\text{Strong} \cap \text{Octet II}) &= 2220 = 60 \cdot 37, \\ T_0(\text{Amino} \cap \text{Octet II}) = T_0(\text{Weak} \cap \text{Octet II}) &= 1998 = 54 \cdot 37, \\ T_0(\text{Octet II}) &= 4218 = 114 \cdot 37. \end{aligned}$$

The intersections of the Purine and Pyrimidine groups with Octet II consist of branches of a single type, so the branch deviations do not cancel but double relative to the sums of the corresponding multiples of 37.

Each 16-codon half of a chemical axis in Octet I has $T = 1850 = 50 \cdot 37$ (Section 3.1.2). Adding the corresponding half carries the divisibility over to the complete groups of the Keto/Amino and Strong/Weak axes, and adding the two octets carries it over to the whole table:

$$\begin{aligned} T_0(\text{Keto}) = T_0(\text{Strong}) &= 1850 + 2220 = 4070 = 110 \cdot 37, \\ T_0(\text{Amino}) = T_0(\text{Weak}) &= 1850 + 1998 = 3848 = 104 \cdot 37, \\ T_0(\text{All}) &= 3700 + 4218 = 7918 = 214 \cdot 37. \end{aligned}$$

The divisibilities listed follow from the mirror deviations $+13/-13$ between the branches of Octet II and from the divisibility of Octet I, and do not form an independent set of results.

The divisibilities $T(\text{Octet I}) = 3700$, $T_0(\text{Octet II}) = 4218$, and $T_0(\text{Keto} \cap \text{Octet II}) = T_0(\text{Strong} \cap \text{Octet II}) = 2220$ correspond, in the group formalism adopted here, to results of Shcherbak and Makukov [8] and Panov and Filatov [5]; see Section 4.4 for further discussion. The complete values of the group quantities under Key 0 are given in `key0_nucleon-data.csv`.

3.2.2 Alignment of Amino and Weak with the lattice of step 37

Among the six 32-codon groups of the three chemical axes, only Amino (id 4) and Weak (id 5) have all four nucleon quantities divisible by 37:

$$\begin{aligned} T_0(\text{Amino}) = T_0(\text{Weak}) &= 3848 = 104 \cdot 37, \\ P_0(\text{Amino}) = P_0(\text{Weak}) &= 2072 = 56 \cdot 37, \\ N_0(\text{Amino}) = N_0(\text{Weak}) &= 1776 = 48 \cdot 37, \\ \Delta_0(\text{Amino}) = \Delta_0(\text{Weak}) &= 296 = 8 \cdot 37. \end{aligned}$$

The divisibility of T was obtained above. In Octet I the halves of both groups have residues $+1$ for P and -1 for N (Section 3.1.2), and in Octet II the opposite:

$$\begin{aligned} P(\text{Amino} \cap \text{Octet I}) = P(\text{Weak} \cap \text{Octet I}) &= 1000 = 27 \cdot 37 + 1, \\ P_0(\text{Amino} \cap \text{Octet II}) = P_0(\text{Weak} \cap \text{Octet II}) &= 1072 = 29 \cdot 37 - 1, \\ N(\text{Amino} \cap \text{Octet I}) = N(\text{Weak} \cap \text{Octet I}) &= 850 = 23 \cdot 37 - 1, \\ N_0(\text{Amino} \cap \text{Octet II}) = N_0(\text{Weak} \cap \text{Octet II}) &= 926 = 25 \cdot 37 + 1. \end{aligned}$$

When the halves are added, these residues cancel each other, and with them the residue of the difference $\Delta = P - N$.

For each group $G \in \{\text{Amino}, \text{Weak}\}$, the ratios to the nucleon sum under Key 0 form a profile with a common denominator of 13:

$$\frac{P_0(G)}{T_0(G)} = \frac{7}{13}, \qquad \frac{N_0(G)}{T_0(G)} = \frac{6}{13}, \qquad \frac{\Delta_0(G)}{T_0(G)} = \frac{1}{13}.$$

3.2.3 One-unit deviations in Δ and the {G} branch

In Octet I, all four branches have $\Delta = 75 = 2 \cdot 37 + 1$ (Section 3.1.2). In Octet II, the deviation has the opposite sign and, under Key 0, occurs in three of the four branches:

$$\begin{aligned} \Delta(\{\text{C}\} \cap \text{Octet II}) = \Delta(\{\text{T}\} \cap \text{Octet II}) = \Delta_0(\{\text{A}\} \cap \text{Octet II}) &= 73 = 2 \cdot 37 - 1, \\ \Delta_0(\{\text{G}\} \cap \text{Octet II}) &= 83 = 2 \cdot 37 + 9. \end{aligned}$$

The equality of the {C} and {T} branches follows from pyrimidine synonymy (Section 3.1.3), whereas their equality with the purine branch {A} is an additional numerical fact. Under Key 0, its two restored positions are occupied by stop codons, so the pool value $\Delta = 73$ is preserved while the group size increases to eight codons. The {G} branch already differs in the service-excluded pool, where it has $\Delta = 72$ rather than 73 in the other branches; the ATG contribution changes this value to 83 (Section 3.1.1).

Adding the Octet I branches gives the values of the one-nucleotide groups in the complete table:

$$\Delta(\{\text{C}\}) = \Delta(\{\text{T}\}) = \Delta_0(\{\text{A}\}) = 148 = 4 \cdot 37, \qquad \Delta_0(\{\text{G}\}) = 158 = 4 \cdot 37 + 10.$$

The {G} branch belongs to Keto, Strong, and Purine, whereas the other three one-nucleotide groups lie on the lattice. The residue $+10$ therefore propagates to all three axes and to the complete table:

$$\begin{aligned} \Delta_0(\text{Keto}) = \Delta_0(\text{Strong}) = \Delta_0(\text{Purine}) &= 306 = 8 \cdot 37 + 10, \\ \Delta_0(\text{All}) &= 602 = 16 \cdot 37 + 10. \end{aligned}$$

The deviation of the complete table in Δ, reported at the beginning of this section, is thus localized in the {G} branch.

The residues of P and N at the 16-codon level also follow from the 8-codon pattern (Section 3.2.2). In the halves of the Amino and Weak groups, two residues $\Delta \equiv +1$ in Octet I add to $+2$, whereas two residues $\Delta \equiv -1$ in Octet II add to -2. Because the corresponding sums T are divisible by 37, $2P = T + \Delta$ and $2N = T - \Delta$ give $(P, N) \equiv (+1, -1) \pmod{37}$ in Octet I and $(P, N) \equiv (-1, +1) \pmod{37}$ in Octet II.

3.3 Localization of the Keto/Amino asymmetry and proton parity

The asymmetry between the Keto and Amino branches is visible at both levels considered here. It is shown first for the full table under Key 0, where it reduces to the nucleon quantities of three molecules, and then for the service-excluded pool, where two remain.

3.3.1 Localization of the Keto/Amino asymmetry

For the full table under Key 0, the differences between the nucleon quantities of the Keto and Amino branches are

$$\begin{aligned} T_0(\text{Keto}) - T_0(\text{Amino}) &= 4070 - 3848 = 222 = 6 \cdot 37, \\ P_0(\text{Keto}) - P_0(\text{Amino}) &= 2188 - 2072 = 116, \\ N_0(\text{Keto}) - N_0(\text{Amino}) &= 1882 - 1776 = 106, \\ \Delta_0(\text{Keto}) - \Delta_0(\text{Amino}) &= 306 - 296 = 10. \end{aligned}$$

Reduction at the family level. By pyrimidine synonymy (Section 2.4), in all 16 codon families the codons NNT and NNC encode the same amino acid. In the 13 families that contain no service codons, the products of NNG and NNA also coincide. Therefore, the Keto branch, represented by the pair NNG and NNT, and the Amino branch, represented by the pair NNA and NNC, make equal nucleon contributions regardless of which amino acids the family encodes.

The TA family also makes equal contributions: its pyrimidine pair TAC and TAT consists of two Tyr codons, whereas its purine pair TAG and TAA consists of two stop codons, one in each branch. The latter cancel when these two stop codons receive equal assignments.

The asymmetry remains in two families, and in each of them one codon pair cancels whereas the other does not. In AT, ATT and ATC cancel, encoding Ile in the Keto and Amino branches; the remaining pair is ATG/ATA: the initiator versus the third Ile codon. In TG, TGT and TGC cancel, encoding Cys; the remaining pair is TGG/TGA: a Trp codon versus a stop codon.

Under Key 0, the initiator is counted as the methionine it encodes, whereas the stop codon TGA contributes zero. Therefore, from the two remaining pairs, for $X \in \{T, P, N, \Delta\}$,

$$X_0(\text{Keto}) - X_0(\text{Amino}) = X(\text{Trp}) + X(\text{Met}) - X(\text{Ile}),$$

so the asymmetry of the full table reduces to the nucleon quantities of three molecules; in particular,

$$T_0(\text{Keto}) - T_0(\text{Amino}) = 204 + 149 - 131 = 222 = 6 \cdot 37.$$

Reduction to the Trp–Ile **pair.** The initiator ATG is the only service codon that contributes to these differences under Key 0. In the service-excluded pool, it is removed from the sums together with the three stop codons, and the asymmetry reduces to the nucleon differences of a single molecular pair, Trp–Ile:

$$\begin{aligned} T_{\text{SE}}(\text{Keto}) - T_{\text{SE}}(\text{Amino}) &= 3921 - 3848 = 73, \\ P_{\text{SE}}(\text{Keto}) - P_{\text{SE}}(\text{Amino}) &= 2108 - 2072 = 36, \\ N_{\text{SE}}(\text{Keto}) - N_{\text{SE}}(\text{Amino}) &= 1813 - 1776 = 37, \\ \Delta_{\text{SE}}(\text{Keto}) - \Delta_{\text{SE}}(\text{Amino}) &= 295 - 296 = -1. \end{aligned}$$

Thus, for $X \in \{T, P, N, \Delta\}$,

$$X_{\text{SE}}(\text{Keto}) - X_{\text{SE}}(\text{Amino}) = X(\text{Trp}) - X(\text{Ile}).$$

This reduction is determined by the synonymy structure and the positions of the service codons.

By the twin-group identities (Section 2.4), $Q_{\text{SE}}(\text{Strong}) = Q_{\text{SE}}(\text{Keto})$ and $Q_{\text{SE}}(\text{Weak}) = Q_{\text{SE}}(\text{Amino})$ for $Q \in \{T, P, N, \Delta\}$. Therefore, the Strong/Weak-axis asymmetry is described by the same differences as the Keto/Amino asymmetry and likewise reduces to the Trp–Ile pair.

Local and global scales. For the pair localized above, the corresponding differences follow directly from the nucleon counts of the amino acids:

$$N(\text{Trp}) - N(\text{Ile}) = 96 - 59 = 37, \qquad P(\text{Trp}) - P(\text{Ile}) = 108 - 72 = 36.$$

The proton and neutron differences differ by one unit, corresponding to one hydrogen atom: tryptophan contains 12 hydrogen atoms, whereas isoleucine contains 13.

The same residues modulo 37 were previously established at the global level for the entire service-excluded pool (Section 3.1.1):

	global	local
N	0	0
P	-1	-1

The global and local facts do not follow from one another: the former refer to the entire service-excluded pool, whereas the latter refer to the nucleon profile of the Trp–Ile pair to which the Keto/Amino asymmetry reduces in this pool.

In the following analysis, the local differences enter the balance conditions for the service codons, whereas the global residues enter the divisibility conditions in the analytical derivation of Key 1 (Section 3.4).

Arithmetic rarity and structural uniqueness of the Trp–Ile pair. The neutron and proton differences of this pair are rare within the fixed set of canonical amino acids.

The value $|\delta N| = 37$. Among the 190 pairs of canonical amino acids, the value $|\delta N| = 37$ occurs for only two pairs: Trp–Ile and Trp–Leu. The latter gives the same value because Leu and Ile are isomers with identical nucleon counts, $P = 72$ and $N = 59$; the pair structurally realized through the AT and TG families is Trp–Ile. The complete distribution of $|\delta N|$ over the 190 pairs is provided in `amino_acids_neutron_differences.csv`.

The value $|\delta P| = 36$. Among the 190 pairs of canonical amino acids, the value $|\delta P| = 36$ occurs only for the same two pairs, Trp–Ile and Trp–Leu. The complete distribution of $|\delta P|$ over the 190 pairs is provided in `amino_acids_proton_differences.csv`.

The value $|\delta T| = 73$. For the Trp–Ile pair, this value follows from the proton and neutron differences given above, which have the same sign. Among all 190 pairs of canonical amino acids, however, $|\delta T| = 73$ occurs only for Trp–Ile and Trp–Leu. The complete distribution of $|\delta T|$ is provided in `amino_acids_nucleon_differences.csv`.

3.3.2 Proton parity

Each of the 20 canonical amino acids has an even total proton count: all P values in `amino_acids_nucleons.csv` are even. In their electronic ground states, the neutral molecules of the canonical amino acids are closed-shell systems in which electrons occupy orbitals in pairs; therefore, their total number of electrons is even. Because in a neutral molecule the number of electrons equals the total number of protons in its nuclei, the parity of P follows.

Consequently, proton sums over amino acids and pairwise proton differences are even, and none of these quantities can equal an odd multiple of 37. Thus, $|\delta P(\text{Trp}, \text{Ile})|$ cannot equal 37: the nearest even values to 37 are 36 and 38, and the actual difference for the localized pair is 36. For the same reason, the proton sum of the service-excluded pool cannot equal the next higher multiple $113 \cdot 37$, which is odd (Section 3.1.1).

Parity imposes only a constraint. It determines neither the direction nor the magnitude of the deviation: the fact that both the local difference and the global sum lie one unit below the corresponding multiple of 37 follows from the specific nucleon counts.

3.4 Analytical derivation of Key 1

The results of Sections 3.1 and 3.2 show that whether the arithmetic regularities appear depends on which service codons a group contains.

Octet I and the pyrimidine-restricted groups contain no service codons at all, and they exhibit arithmetic order (Sections 3.1.2 and 3.1.3). The Amino and Weak branches contain stop codons but not the initiator ATG: all four of their nucleon quantities are divisible by 37 already in the service-excluded pool, and the zero contribution of the stop codons leaves them unchanged, so the divisibility is preserved under Key 0 (Section 3.2.2). In the Keto and Strong branches, which contain the initiator ATG, the proton and neutron sums and the difference Δ deviate under Key 0 from the 37-lattice. The deviation in Δ is localized in the {G} branch, which contains ATG: the other three one-nucleotide groups share a single value of Δ, divisible by 37 (Section 3.2.3).

This points to the possibility of a different parametrization of the service codons, one that reconciles the arithmetic regularities of the groups considered. In this section we derive such a parametrization analytically from conditions of symmetry, divisibility, and minimality. The resulting assignment serves as a formal device for reconciling the sums of the complete table; the underlying arithmetic regularities are established in the service-excluded pool. The specificity of the result across natural codes is examined in Section 4.1.

For the assignment K sought, let $(p_{\text{stop}}, n_{\text{stop}})$ denote the common parameters of the three stop codons and $(p_{\text{start}}, n_{\text{start}})$ those of the initiator ATG. All parameters are non-negative integers. The total contribution of the service codons is minimized separately for neutrons and protons, as defined in Section 2.5. We first determine the neutron parameters, then the proton parameters.

3.4.1 Determination of the neutron parameters

Distribution of the service codons. All four service codons belong to Octet II, so the entire added contribution falls in that octet. The initiator ATG and the stop codon TAG have G in the third position and belong to the Keto branch; the stop codons TAA and TGA have A in the third position and belong to the Amino branch. Thus Keto contains one stop codon and the initiator, whereas Amino contains two stop codons. The same distribution of service positions holds for Strong and Weak. By the twin-group identities (Section 2.4), Keto/Amino balance also ensures Strong/Weak balance.

Balance equation for Keto and Amino. The complete neutron sums consist of the service-excluded pool sums (Section 3.3.1) and the unknown contributions:

$$N_K(\text{Keto}) = 1813 + n_{\text{start}} + n_{\text{stop}},$$
$$N_K(\text{Amino}) = 1776 + 2\, n_{\text{stop}}.$$

The condition $N_K(\text{Keto}) = N_K(\text{Amino})$ gives

$$n_{\text{stop}} - n_{\text{start}} = 37.$$

The right-hand side is the difference between the neutron counts of tryptophan and isoleucine, to which the pool imbalance has been reduced. Balance fixes the difference between the parameters, leaving a one-parameter family. In the main derivation, the next condition is divisibility of the total sum. Its compatibility with the choice of chemical axis is examined in Section 3.4.3.

Divisibility of the total neutron sum by 37. The service-excluded pool has $N(\mathcal{C}_{\text{SE}}) = 3589 = 97 \cdot 37$ (Section 3.1.1). Under Key 0, the initiator contributes the neutron count of methionine, giving

$$N_0(\text{All}) = 3589 + 69 = 3658 = 99 \cdot 37 - 5.$$

To preserve divisibility in the complete table, we require

$$N_K(\text{All}) = 3589 + n_{\text{start}} + 3\,n_{\text{stop}} \equiv 0 \pmod{37},$$

or equivalently

$$n_{\text{start}} + 3\,n_{\text{stop}} \equiv 0 \pmod{37}.$$

Solution. Substituting $n_{\text{stop}} = n_{\text{start}} + 37$ from the first condition into the second yields

$$4\,n_{\text{start}} + 111 \equiv 0 \pmod{37}.$$

Since $111 = 3 \cdot 37 \equiv 0 \pmod{37}$ and $\gcd(4, 37) = 1$, this reduces to

$$n_{\text{start}} \equiv 0 \pmod{37},$$

giving the family of non-negative integer solutions $n_{\text{start}} = 0, 37, 74, \ldots$. The minimal solution is

$$n_{\text{start}} = 0, \quad n_{\text{stop}} = 37.$$

The divisibility requirement does not change the minimum just obtained. With non-negative integer parameters, the Keto/Amino balance alone admits $n_{\text{start}} = 0, 1, 2, \ldots$ with $n_{\text{stop}} = n_{\text{start}} + 37$; divisibility narrows this family to $n_{\text{start}} = 0, 37, 74, \ldots$, but in both cases the minimal total added contribution is attained at $n_{\text{start}} = 0$ and $n_{\text{stop}} = 37$. For the proton component the divisibility requirement does change the minimum. The structure of the solution families and the role of each condition are addressed in Section 3.4.3.

Inputs and the result of the derivation. The system uses the local difference 37 and the total neutron sum of the pool, $3589 = 97 \cdot 37$. Both quantities are established before assigning values to the service codons. The coefficient 4 in the combined congruence arises when the balance equation is substituted into the sum of the contributions of three stop codons and the initiator. Coprimality of 4 and 37 allows the congruence to be reduced; this is a property of the equation, and the parameters obtained constitute its minimal solution.

Coincidence with the Octet I sum. After determining the parameters, the total neutron sum is

$$N_1(\text{All}) = 3589 + 0 + 3 \cdot 37 = 3700 = T(\text{Octet I}).$$

Equality with the nucleon sum of Octet I was not used as a condition of the derivation. It already holds at the minimal neutron assignment satisfying balance alone. At the level of Octet II, the same relation takes the form

$$N_1(\text{Octet II}) = 1889 + 0 + 3 \cdot 37 = 2000 = P(\text{Octet I}).$$

This restates the preceding equality after subtracting $N(\text{Octet I}) = 1700$, rather than constituting a separate coincidence. Balance and the twin-group identities give

$$\begin{aligned} N_1(\text{Keto}) = N_1(\text{Amino}) = N_1(\text{Strong}) = N_1(\text{Weak}) \\ = 1850 = \tfrac{1}{2}\,T(\text{Octet I}). \end{aligned}$$

Here 1850 is obtained by halving the total sum just found. Further consequences are considered in Section 3.5.

3.4.2 Determination of the proton parameters

The proton parameters are determined separately from the neutron parameters under the same conditions: Keto/Amino balance, divisibility of the total sum by 37, and minimality. Equality of the proton and neutron stop parameters is not assumed.

Balance equation for Keto and Amino. From the service-excluded pool sums (Section 3.3.1), we obtain

$$P_K(\text{Keto}) = 2108 + p_{\text{start}} + p_{\text{stop}},$$
$$P_K(\text{Amino}) = 2072 + 2\,p_{\text{stop}}.$$

The condition $P_K(\text{Keto}) = P_K(\text{Amino})$ gives

$$p_{\text{stop}} - p_{\text{start}} = 36.$$

The right-hand side equals $P(\text{Trp}) - P(\text{Ile})$. Parity rules out 37 but does not determine the actual difference 36, which follows from the proton counts of these molecules (Section 3.3.2).

Divisibility of the total proton sum by 37. The proton sum of the service-excluded pool is

$$\begin{aligned} P(\mathcal{C}_{\text{SE}}) &= P(\text{Octet I}) + P_{\text{SE}}(\text{Octet II}) \\ &= 2000 + 2180 = 4180 = 113 \cdot 37 - 1. \end{aligned}$$

This is a one-unit deviation from the nearest multiple of 37. For the complete table, we require

$$P_K(\text{All}) = 4180 + p_{\text{start}} + 3\,p_{\text{stop}} \equiv 0 \pmod{37},$$

or equivalently

$$p_{\text{start}} + 3\,p_{\text{stop}} \equiv 1 \pmod{37}.$$

Solution. Substituting $p_{\text{stop}} = p_{\text{start}} + 36$ gives

$$4\,p_{\text{start}} + 108 \equiv 1 \pmod{37}.$$

Since $108 = 3 \cdot 37 - 3$ and $\gcd(4, 37) = 1$,

$$4\,p_{\text{start}} \equiv 4 \pmod{37}, \qquad p_{\text{start}} \equiv 1 \pmod{37}.$$

The non-negative integer solutions have $p_{\text{start}} = 1, 38, 75, \ldots$. The minimal solution is

$$p_{\text{start}} = 1, \qquad p_{\text{stop}} = 37.$$

The family structure and the minimality of this choice are considered in Section 3.4.3.

The resulting equality of the stop parameters. The proton system uses the local difference 36 and the one-unit deviation of the total pool sum from a multiple of 37. As in the neutron derivation, the coefficient 4 is determined by the distribution of service positions and the balance equation. Reducing the congruence and selecting the minimum yields parameters that can be compared with the neutron solution (Section 3.4.1):

$$p_{\text{stop}} = n_{\text{stop}} = 37.$$

Equality of the proton and neutron parameters was not among the conditions of the two systems. It arises when their minimal non-negative integer solutions are selected.

Macroscopic consequences. Under Key 1, the total proton sum is 4292, while balance and the twin-group identities give

$$\begin{aligned} P_1(\text{Keto}) = P_1(\text{Amino}) = P_1(\text{Strong}) = P_1(\text{Weak}) \\ = 2146 = \tfrac{1}{2}\, T_1(\text{Octet II}). \end{aligned}$$

The corresponding complete-table equality is

$$P_1(\text{All}) = T_1(\text{Octet II}) = 4292.$$

The nucleon sum of Octet II here is computed under Key 1 and depends on the service-codon assignment. This equality follows from $N_1(\text{Octet II}) = P(\text{Octet I})$ and the definition $T = P + N$: it expresses the same relation between the complete table and the octets as the neutron coincidence. The full analysis is given in Section 3.5.

3.4.3 Hierarchy of constraints and the choice of Key 1

The systems in Sections 3.4.1 and 3.4.2 share the same construction. For non-negative integer parameters, the assignment is selected in three steps:

1. Keto/Amino balance fixes the differences $n_{\text{stop}} - n_{\text{start}} = 37$ and $p_{\text{stop}} - p_{\text{start}} = 36$. Two free parameters remain: the neutron and proton values assigned to the initiator. Every member of this family equalizes the four 32-codon groups Keto, Amino, Strong, and Weak in P and N, and hence in T and Δ.

2. Divisibility of the total sums by 37 restricts the initiator values to $n_{\text{start}} = 37k$ and $p_{\text{start}} = 1 + 37m$, where k, m are non-negative integers. The family still has two free integer parameters, but the step in each direction increases to 37.

3. The total added contributions are $111 + 148k$ for neutrons and $112 + 148m$ for protons. They increase strictly with k and m, respectively, so the unique minimum is attained at $k = m = 0$, giving Key 1: $(p_{\text{stop}}, n_{\text{stop}}) = (37, 37)$, $(p_{\text{start}}, n_{\text{start}}) = (1, 0)$.

Uniqueness refers to the minimum under the stated conditions: balance and divisibility also admit larger assignments. In particular, equality of the proton and neutron stop parameters holds whenever $k = m$, but their common value equals 37 only at $k = m = 0$.

The contribution of the divisibility condition. We compare the assignments with the minimal total added contribution, obtained separately for protons and for neutrons, under the Keto/Amino balance alone and under the additional requirement of divisibility by 37; in both cases the parameters are non-negative integers:

$$\begin{aligned} \text{balance only:} \quad & (p_{\text{stop}}, n_{\text{stop}}) = (36, 37), \quad (p_{\text{start}}, n_{\text{start}}) = (0, 0), \\ \text{balance and divisibility:} \quad & (p_{\text{stop}}, n_{\text{stop}}) = (37, 37), \quad (p_{\text{start}}, n_{\text{start}}) = (1, 0). \end{aligned}$$

The divisibility requirement does not change the minimal neutron assignment: the neutron sum of the complete table equals 3700 and is divisible by 37 under the balance alone. Under the balance alone the minimal proton sum is $4288 = 116 \cdot 37 - 4$; adding one proton to each of the four service positions raises it to $4292 = 116 \cdot 37$. The balance is preserved, since each half of the Keto/Amino axis contains two service positions. The additional divisibility requirement therefore increases the minimal contribution by four protons and requires no additional neutrons.

The same increment removes the difference in Δ across the Purine/Pyrimidine axis. For the Pyrimidine branch $\Delta = 296$ is independent of the parametrization (Section 3.1.3), whereas under the minimal assignment that secures the Keto/Amino balance alone the Purine branch has $\Delta = 292$: each stop codon carries $\Delta = 36 - 37 = -1$, which shifts the service-excluded pool value

295 (Section 3.1.1) down by three units. All four service codons belong to the Purine branch, so adding one proton to each of them equalizes the values of the two branches. An alternative derivation is considered below, in which this equality is imposed together with the Keto/Amino balance and the divisibility requirement is not introduced.

The stop-parameter equality as a two-axis condition. Instead of divisibility, one can require equal Δ values for Purine and Pyrimidine. Denote the differences of the service-codon parameters by $\Delta_{\text{stop}} = p_{\text{stop}} - n_{\text{stop}}$ and $\Delta_{\text{start}} = p_{\text{start}} - n_{\text{start}}$. Keto/Amino balance gives

$$\Delta_{\text{start}} = \Delta_{\text{stop}} + 1.$$

All service codons belong to Purine, and the difference between the Δ values of the two branches in the service-excluded pool is $\Delta_{\text{SE}}(\text{Pyrimidine}) - \Delta_{\text{SE}}(\text{Purine}) = 1$. The additional balance therefore requires

$$\Delta_{\text{start}} + 3\,\Delta_{\text{stop}} = 1.$$

Together these equations give $\Delta_{\text{stop}} = 0$ and $\Delta_{\text{start}} = 1$. Thus $p_{\text{stop}} = n_{\text{stop}}$ and $p_{\text{start}} - n_{\text{start}} = 1$ are obtained without requiring divisibility.

Together with Keto/Amino balance, this leaves the one-parameter family $(p_{\text{stop}}, n_{\text{stop}}) = (37+t, 37+t)$, $(p_{\text{start}}, n_{\text{start}}) = (1+t, t)$ for integer $t \geq 0$. Along this family the total neutron sum is $3700 + 4t$ and the total proton sum is $4292 + 4t$. The minimal total contribution is attained at $t = 0$, again giving Key 1. Both sums are divisible by 37 only when $t \equiv 0 \pmod{37}$; at the minimal assignment, divisibility is a consequence rather than a condition of this derivation. Equality of the stop parameters here depends on the agreement of the differences across two axes: $\delta P(\text{Trp}, \text{Ile}) - \delta N(\text{Trp}, \text{Ile}) = -1$ (Section 3.3.1) and the difference in Δ between Pyrimidine and Purine given above.

Why the balance in P and N is imposed on Keto/Amino. Equal proton and neutron sums for Purine and Pyrimidine would require different service-codon contributions:

$$p_{\text{start}} + 3\,p_{\text{stop}} = 212, \qquad n_{\text{start}} + 3\,n_{\text{stop}} = 211.$$

However, $212 \equiv -10 \pmod{37}$ and $211 \equiv -11 \pmod{37}$, whereas divisibility of the total sums requires residues 1 and 0, respectively. Such a balance is therefore incompatible with divisibility by 37. Among the three third-position axes, this requirement is compatible with balance in P and N on Keto/Amino and its structural copy on Strong/Weak. Balance in Δ on Purine/Pyrimidine does not impose these two separate equalities and is compatible with the assignment obtained.

The common basis and the additional condition. Both derivations use equal assignments to the three stop codons, the same initial sums, Keto/Amino balance, non-negative integer parameters, and minimality of the total contribution. This basis alone already gives a unique minimum: as shown above, it coincides with Key 1 in neutrons and is one proton lower at each service position. An additional condition narrows the set of admissible assignments, and the minimum is sought again within the narrowed set. With separate minimization of the proton and neutron contributions, the new minimum is Key 1 if and only if the condition admits Key 1 and excludes all assignments with $p_{\text{start}} = 0$. Both conditions considered above are of this kind: divisibility leaves $p_{\text{start}} = 1, 38, 75, \ldots$, and equality of Δ between Purine and Pyrimidine gives $p_{\text{start}} = n_{\text{start}} + 1$. So is the requirement that Δ be non-negative for all service-codon assignments, $\Delta_{\text{start}} \geq 0$ and $\Delta_{\text{stop}} \geq 0$: given $\Delta_{\text{start}} = \Delta_{\text{stop}} + 1$, it reduces to $p_{\text{start}} \geq n_{\text{start}} + 1$. These examples do not exhaust such conditions. Divisibility of the nucleon sum T alone, by contrast, does not meet the criterion: it also admits the assignment $(p_{\text{start}}, n_{\text{start}}) = (0, 1)$, $(p_{\text{stop}}, n_{\text{stop}}) = (36, 38)$, and the smallest proton and neutron contributions are then reached at different assignments. The main derivation therefore requires divisibility separately for P and N.

The coincidence of the minima under different conditions does not make these conditions equivalent and is not an independent confirmation of Key 1. It follows from the properties of the service-excluded pool established above. Non-negativity of Δ leads to Key 1 because of the difference $\delta P(\text{Trp}, \text{Ile}) - \delta N(\text{Trp}, \text{Ile}) = -1$, from which the balance gives $\Delta_{\text{start}} = \Delta_{\text{stop}} + 1$; divisibility does so because the pool sums and the differences of the tryptophan–isoleucine pair have the same signed residues modulo 37, -1 for P and 0 for N (Section 3.3.1); equality of Δ between Purine and Pyrimidine does so because of the agreement of the differences across two axes given above. Under non-negativity of Δ and under equality of Δ between Purine and Pyrimidine, divisibility of the total sums by 37 is not imposed but holds at the minimum obtained. The main derivation uses divisibility because Key 1 is constructed to reconcile the sums of the complete table with the primary arithmetic criterion of the analysis, divisibility by 37 (Section 2.6). The choice of the additional condition is determined by this aim, not by the number of conditions that lead to the same minimum.

The resulting value $1850 = \frac{1}{2}\,T(\text{Octet I})$ allows the neutron assignments to be checked in reverse: $1850 - 1776 = 74 = 2 \cdot 37$ is assigned to the two stop codons in Amino, and $1850 - 1813 = 37$ to the stop codon and initiator in Keto. This again gives $n_{\text{stop}} = 37$ and $n_{\text{start}} = 0$. This check uses equality with half the Octet I sum after it has been obtained; it is not an independent basis for the derivation.

3.5 Consequences under Key 1

Under Key 1, Keto/Amino balance extends to four 32-codon groups, while equality of the total neutron sum and the nucleon sum of Octet I connects several levels of partitioning. This section examines these consequences, the divisibility properties of the resulting sums, and the residual Purine/Pyrimidine asymmetry. An overview of the groups under Key 1 is given in Figure 4.

3.5.1 Alignment of the chemical-axis 32-codon groups

Under Key 1, the Keto, Amino, Strong, and Weak groups have identical values of the four nucleon quantities. For each of these groups G:

$$T_1(G) = 3996, \quad P_1(G) = 2146,$$
$$N_1(G) = 1850, \quad \Delta_1(G) = 296.$$

Keto/Amino balance, together with the twin-group identities, ensures equality of the sums of all four groups in P and N, while the definitions $T = P + N$ and $\Delta = P - N$ ensure equality in T and Δ. Thus the four aligned groups are a consequence of the chosen assignment, rather than four separate confirmations. Their sums are half those of the complete table. All group values are given in `key1_nucleon-data.csv`.

Purine and Pyrimidine have the same Δ but differ in P, N, and T. This difference is examined in Section 3.5.5.

3.5.2 Macroscopic identities and propagation to chemical-axis halves

The relation between the complete table and the octets has three equivalent forms:

$$N_1(\text{All}) = T(\text{Octet I}) = 3700 = 100 \cdot 37,$$
$$P_1(\text{All}) = T_1(\text{Octet II}) = 4292 = 116 \cdot 37,$$
$$N_1(\text{Octet II}) = P(\text{Octet I}) = 2000 = 2 \cdot 1000.$$

This is one relation, obtained in Section 3.4.1 after determining the parameters. Subtracting $N(\text{Octet I}) = 1700$ from the first equality gives the third. Adding $P_1(\text{Octet II})$ to the third gives



Figure 4: Nucleon sums of codon groups under Key 1: each stop codon is assigned $(P, N) = (37, 37)$, and the initiator ATG is assigned $(P, N) = (1, 0)$. The group layout and divisibility color code are the same as in Figure 2; the comparison with Key 0 is shown in Figure 3. Generated by the accompanying `visualization.html`.

the second, since $T = P + N$. The transformations also hold in reverse. The sums of Octet I are independent of parametrization; those of Octet II include the service-codon assignments.

For each of the four aligned groups G—Keto, Amino, Strong, and Weak—this relation is reproduced at the level of its octet intersections:

$$\begin{aligned} N_1(G) &= T(G \cap \text{Octet I}) = 1850, \\ P_1(G) &= T_1(G \cap \text{Octet II}) = 2146, \\ N_1(G \cap \text{Octet II}) &= P(G \cap \text{Octet I}) = 1000. \end{aligned}$$

Each intersection contains 16 codons. In Octet I, each such group includes two codons from every four-fold degenerate family and therefore carries half the octet sums. In Octet II, equality of the halves along the two axes follows from complete-table balance after subtracting the equal contributions of Octet I. The equalities above are therefore half-scale forms of the same relation. This alignment of Octet II does not hold for Purine and Pyrimidine. The complete list of equalities is given in `key1_equalities.csv`.

Rational profile. The complete table and the four aligned groups share the same ratios of nucleon quantities under Key 1. Denoting any of these groups by G, we obtain:

$$\frac{P_1(G)}{T_1(G)} = \frac{29}{54}, \qquad \frac{N_1(G)}{T_1(G)} = \frac{25}{54}, \qquad \frac{\Delta_1(G)}{T_1(G)} = \frac{2}{27}.$$

Accordingly, $P_1(G) : N_1(G) = 29 : 25$. This is a normalized expression of the same sums. For each of the four groups, $2146 = 29 \cdot 2 \cdot 37$ is both the proton sum of the group and the nucleon sum of its intersection with Octet II; $1850 = 25 \cdot 2 \cdot 37$ links the neutron sum to the intersection with Octet I. The ratio profiles of all groups are given in `key1_ratios.csv`.

3.5.3 Divisibility by 37 and by 999

Divisibility by 37. In the main derivation of Key 1, divisibility of the total sums in P and N is explicitly imposed. For neutrons it already holds at the minimal assignment satisfying balance alone; for protons it changes the minimum (Section 3.4.3). The resulting sums are

$$N_1(\text{All}) = 100 \cdot 37, \qquad P_1(\text{All}) = 116 \cdot 37.$$

Balance divides each sum into two equal integer parts. Since 37 is odd, both parts are also divisible by 37. For each of the four aligned groups G, we have

$$\begin{aligned} T_1(G) &= 108 \cdot 37, \quad P_1(G) = 58 \cdot 37, \\ N_1(G) &= 50 \cdot 37, \quad \Delta_1(G) = 8 \cdot 37. \end{aligned}$$

Divisibility of T and Δ follows by adding and subtracting P and N; in particular, $\Delta_1(\text{All}) = 4292 - 3700 = 592 = 16 \cdot 37$.

A more general property holds for the nucleon sum:

$$T_1(G) \equiv T_0(G) \pmod{37}$$

for any codon group G. In the transition from Key 0 to Key 1, the contribution of each stop codon to T increases by $74 = 2 \cdot 37$, whereas that of ATG changes by $1 - 149 = -148 = -4 \cdot 37$. Divisibility of T by 37 is therefore preserved in this transition regardless of the group's composition. The divisibility records under Key 1 are given in `key1_divisibility-37.csv`.

Divisibility by 999. The total nucleon sum has the form

$$T_1(\text{All}) = 7992 = 8 \cdot 999 = 6^3 \cdot 37.$$

Unlike divisibility by 37, divisibility by $999 = 27 \cdot 37$ was not imposed as a separate condition. It characterizes the minimal solution obtained. The nucleon sum of each of the four aligned halves under Key 1 is $3996 = 4 \cdot 999$; this follows by halving the total sum.

3.5.4 The Δ-invariant across all six 32-codon groups

Under Key 1, all six third-position 32-codon groups have the same proton-neutron difference:

$$\begin{aligned}\Delta_1(\text{Keto}) = \Delta_1(\text{Amino}) = \Delta_1(\text{Strong}) = \Delta_1(\text{Weak}) \\ = \Delta_1(\text{Purine}) = \Delta(\text{Pyrimidine}) = 296.\end{aligned}$$

For the first four groups, equality follows from balance in P and N. Equality of Purine and Pyrimidine in Δ is not imposed separately in the main derivation; in the alternative two-axis derivation it serves as a condition in place of divisibility (Section 3.4.3). Its role therefore differs between the two derivations.

The common difference is reproduced at lower levels of partitioning. Every eight-codon group of Octet I selected by a single nucleotide in the third position has $\Delta = 75$ independently of parametrization (Section 3.1.2). Under Key 1, every corresponding group of Octet II has $\Delta = 73$:

$$\Delta(G_X \cap \text{Octet I}) = 75, \qquad \Delta_1(G_X \cap \text{Octet II}) = 73 \qquad (|X| = 1).$$

In the pyrimidine groups, 73 is also independent of parametrization (Section 3.1.3); in the purine groups it is attained under the derived assignment. Adding two eight-codon groups within each octet gives

$$\Delta(G_X \cap \text{Octet I}) = 150, \qquad \Delta_1(G_X \cap \text{Octet II}) = 146 \qquad (|X| = 2).$$

Adding across the octets gives, for each one-nucleotide group of the complete table,

$$\Delta_1(G_X) = 75 + 73 = 148 = 4 \cdot 37 \qquad (|X| = 1).$$

Under Key 0, the {G} group had $\Delta_0(\{\text{G}\}) = 158$, whereas the other three already had 148 (Section 3.2.3). Under Key 1, all four values coincide. Both routes of addition reach the 32-codon level:

$$296 = 2 \cdot 148 = 150 + 146.$$

Thus the cascade describes one agreement at several levels. It also persists for nonminimal assignments in the two-axis family of Section 3.4.3: increasing both parameters by the same amount leaves Δ unchanged for every service codon.

3.5.5 Purine/Pyrimidine asymmetry

Under Key 1, the Purine/Pyrimidine axis retains the following differences between the sums of its halves:

$$\begin{aligned}N(\text{Pyrimidine}) - N_1(\text{Purine}) &= 100, \\ P(\text{Pyrimidine}) - P_1(\text{Purine}) &= 100, \\ T(\text{Pyrimidine}) - T_1(\text{Purine}) &= 200, \\ \Delta(\text{Pyrimidine}) - \Delta_1(\text{Purine}) &= 0.\end{aligned}$$

The entire asymmetry lies in Octet II: in Octet I, the purine and pyrimidine halves have equal sums because its families are four-fold degenerate.

In the service-excluded pool, the Pyrimidine–Purine differences are 212 in P and 211 in N (Section 3.1.1). Key 1 adds 112 protons and 111 neutrons to the purine half, so

$$212 - 112 = 211 - 111 = 100.$$

Equality of the two shifts is equivalent to equality of Δ between Purine and Pyrimidine. Their common magnitude of 100 is obtained from the initial sums after assigning the parameters; it was not imposed as a condition of the derivation.

Equalities at the one-nucleotide level. Equality of the sums of the {C} and {T} groups is independent of parametrization and follows from pyrimidine synonymy (Section 2.4). Since Keto combines {G} and {T}, whereas Amino combines {A} and {C}, balance in P and N gives

$$P_1(\{\mathrm{G}\}) = P_1(\{\mathrm{A}\}), \qquad N_1(\{\mathrm{G}\}) = N_1(\{\mathrm{A}\}).$$

This restates balance and holds for any assignment that satisfies it. Under Key 1, the two pairs of groups have the values

$$\big(T(\{\mathrm{C}\}), P(\{\mathrm{C}\}), N(\{\mathrm{C}\})\big) = (2048, 1098, 950),$$
$$\big(T_1(\{\mathrm{G}\}), P_1(\{\mathrm{G}\}), N_1(\{\mathrm{G}\})\big) = (1948, 1048, 900),$$

with the same values for {T} and {A}, respectively. The difference between the pairs is 50 in P and N and 100 in T; doubling reproduces the differences between the two 32-codon halves. After subtracting the equal Octet I contributions, each of the groups $\{\mathrm{G}\} \cap$ Octet II and $\{\mathrm{A}\} \cap$ Octet II under Key 1 has a nucleon sum of 1023, a proton sum of 548, and a neutron sum of 475.

Inside Octet II at the 16-codon level. The purine and pyrimidine halves of Octet II have neutron sums of 950 and 1050 and proton sums of 1096 and 1196, respectively. They deviate by ± 50 from the mean values: 1000 for the neutron sum and 1146 for the proton sum. These are the sums attained under Key 1 by the halves of this octet along the Keto/Amino and Strong/Weak axes. This is another expression of the same asymmetry: the two halves lie symmetrically about the mean but are not equal to each other.

The limit of alignment. The residual asymmetry does not imply that complete balance across the three axes is impossible. In the two-axis family of Section 3.4.3, where the stop codons have parameters $(37 + t, 37 + t)$ and the initiator has $(1 + t, t)$, both Pyrimidine–Purine shifts equal $100 - 4t$. At $t = 25$, the assignments $(62, 62)$ to the stop codons and $(26, 25)$ to the initiator equalize all six halves in P and N. However, the total sums become $P = 4392$ and $N = 3800$ and are not divisible by 37; the neutron equality with $T(\text{Octet I}) = 3700$ is also lost. Key 1 fixes the minimal reconciliation of balance with divisibility, retaining this asymmetry.

3.6 Synthesis

The observed regularities connect two levels of organization of the standard genetic code: the structure of codon groups and the nucleon composition of the encoded amino acids. The partitions are based on the chemical properties of bases, pyrimidine synonymy, and Rumer's octets. In the adopted representation of free neutral amino acids, proton and neutron counts are uniquely determined by molecular composition, while group sums account for the number of codons encoding each amino acid. The choice of the third position, molecular representation, and modulus 37 defines the analytical framework; these choices alone do not entail the observed numerical values.

The structural partitions exhibit arithmetic agreement even before values are assigned to service codons. In Octet I, the proton and neutron sums are 2000 and 1700, and their sum,

$3700 = 100 \cdot 37$, connects a decimal scale with modulus 37. The octet's structure ensures exact scaling of these quantities across its groups but does not determine their initial values. For the entire service-excluded pool, the neutron sum is $3589 = 97 \cdot 37$, while the proton sum is $4180 = 113 \cdot 37 - 1$ (Section 3.1).

The local asymmetry agrees with this global pattern. The difference between Keto and Amino reduces to the tryptophan–isoleucine pair, whose differences are 37 in neutrons and 36 in protons. Among the 190 pairs of canonical amino acids, this combination of differences occurs only for tryptophan–isoleucine and tryptophan–leucine. The agreement is explained by isoleucine and leucine being isomers with identical nucleon counts (Section 3.3.1). The even proton counts of all canonical amino acids rule out a proton difference of 37 but do not determine how far the actual difference must deviate from it. The value 36 is the nearest permitted value below 37, and the proton and neutron imbalances differ by exactly one unit (Section 3.3.2).

One-unit deviations occur at different levels of the structure and have different arithmetic origins. In Octet I, the values $75 = 2 \cdot 37 + 1$ arise by dividing its difference $\Delta = 300$ among four equal eight-codon groups. In the pyrimidine groups of Octet II, the value is displaced in the opposite direction: $73 = 2 \cdot 37 - 1$. When the octets are combined, the deviations cancel: $75 + 73 = 148 = 4 \cdot 37$. In the service-excluded pool, the {A} branch of Octet II also has $\Delta = 73$, an equality that does not follow from pyrimidine synonymy. Only the {G} branch differs, with a value of 72; the excluded ATG position lies in this branch (Section 3.2.3). A further agreement appears under Key 0: methionine's total nucleon count, $149 = 4 \cdot 37 + 1$, compensates for the remainder of -1 in the pool's nucleon sum. Thus, one-unit deviations have several sources, and their recurrences propagate through structural relations among groups.

The arrangement of service codons allows local balance to be reconciled with global sums. Keto contains one stop codon and ATG, whereas Amino contains two stop codons. With the same assignment to all stop codons, the minimal non-negative integer compensation for the neutron imbalance assigns 37 neutrons to each stop codon and zero to ATG. This gives

$$N_1(\text{All}) = 3589 + 3 \cdot 37 = 3700 = T(\text{Octet I}).$$

The coincidence with the Octet I sum was not used to select the compensation. Here, the magnitude of the local imbalance, the number and arrangement of stop codons, and the sum of structural Octet I agree.

On the proton side, moving from the minimal balanced assignment to the minimal balanced assignment with divisibility by 37 requires four additional protons — one for each service position. Their equal distribution between Keto and Amino preserves balance. Since all four positions belong to Purine, the same increment eliminates the difference in Δ between Purine and Pyrimidine while preserving the original pyrimidine sums. The resulting assignment is Key 1: $(P, N) = (37, 37)$ for each stop codon and $(1, 0)$ for ATG (Section 3.4). Its simplicity expresses the compatibility of the initial numbers with the arrangement of service positions; uniqueness is relative to the adopted conditions and minimality criterion.

The resulting assignment brings these relations together in the complete table. Several macroscopic equalities express a single relation with Octet I; equality of Δ is compatible with Pyrimidine exceeding Purine by 100 in both proton and neutron sums. The nucleon sum $7992 = 8 \cdot 999$ adds to the numerical profile, although divisibility by 999 was not imposed as a separate condition (Section 3.5). The result concerns the agreement among molecular counts, structural partitions, and formal assignments; repeated manifestations of one relation do not constitute separate pieces of evidence. The specificity of this agreement is assessed next through comparison with alternative codes, deterministic controls, and a null model (Section 4).

4 Discussion

This section examines how far the regularities reported above are specific to the standard code and its structure, rather than artifacts of the analysis. Section 4.1 compares the standard code with alternative NCBI codes in terms of the neutron deficit and proton residue of the service-excluded pool, the localization of the asymmetry in the tryptophan–isoleucine pair, and the results of re-deriving the service-codon assignments; Section 4.2 asks whether the third codon position and the modulus 37 are distinguished from other positions and divisors; and Section 4.3 assesses the concentration on the 37-lattice against a random-code null. Section 4.4 situates the results relative to prior work, and Section 4.5 states the limitations and outlook.

4.1 Specificity across alternative genetic codes

The comparison of the 27 translation tables in the NCBI registry shows which arithmetic properties of the standard code occur in other genetic codes. For the service-excluded pool we compare the neutron deficit and the proton residue (Section 3.1.1), together with the differences of the neutron and proton sums between Keto and Amino. For the tables that reproduce the standard values of these differences, we check whether their localization in the tryptophan–isoleucine pair is preserved (Section 3.3.1). Separately, the same conditions of balance, divisibility and minimality are applied to each table (Section 3.4), and the resulting assignments to the service codons and the group sums are compared.

The same two models are used in all comparisons. Model 1 excludes ATG and every codon listed in the registry as a stop codon; Model 2 excludes ATG and only the permanent stop codons. Context-dependent stop codons are included in Model 2 with the amino acid assignments recorded for them. The distinction affects tables 27, 28 and 31. Excluding ATG alone while retaining the elongation assignments of the alternative start codons is a convention shared by the two models (Section 2.2).

Neutron deficit and proton residue. For the standard code the neutron sum of the service-excluded pool falls short of the nucleon sum of Octet I by $3700 - 3589 = 111 = 3 \cdot 37$, and the proton sum has residue -1 modulo 37. The first test checks whether these two values are reproduced in the other tables.

In `deficit_models_analysis.csv` the neutron deficit is computed relative to three reference sums: the constant 3700 of the standard code; the nucleon sum of structural Octet I, that is, of the same eight codon families as in the standard code but under the amino acid assignments of the table tested; and the nucleon sum of functional Octet I, that is, of all fully four-fold degenerate families of the table itself (Section 2.7). All three deficits are determined for each of the two pool models, which gives six variants of the comparison. The proton residue is taken modulo 37 directly from the proton sum of the service-excluded pool, separately for each model, and requires no reference sum.

The neutron deficit 111 under any of the three reference sums and the proton residue -1 modulo 37 each select the same tables: 1, 11 and 28 under Model 1 and only 1 and 11 under Model 2. In the remaining 24 tables neither of the two features is reproduced under either model.

Table 11 has the same amino acid assignments and stop codons as the standard table. In table 28 the exclusion of three context-dependent stop codons under Model 1 leaves the same pool; their return under Model 2 breaks both matches. The matching rows therefore represent one and the same arithmetic of the original pool rather than independent repetitions of the result.

Preservation of the Keto/Amino asymmetry and its localization. The second test checks whether the differences of the neutron and proton sums, Keto minus Amino, are reproduced; in the standard code they equal 37 and 36 respectively. The values for all tables and both models are given in `keto_amino_balance_models.csv`. This pair of differences is reproduced in tables 1,

6, 11, 29 and 30 under both models, and in tables 27 and 28 only under Model 1. Agreement of the differences does not by itself establish their localization; its preservation is checked against the codon assignments.

In tables 6, 29 and 30 the codons TAA and TAG are reassigned to one and the same amino acid: glutamine, tyrosine or glutamic acid respectively. Within the TA family the codon TAG belongs to the Keto branch and TAA to the Amino branch, so the added contributions cancel in the difference. There are no other changes in these tables: TGG and all three isoleucine codons retain their standard assignments, and the localization in the tryptophan–isoleucine pair is preserved. The same mechanism operates in table 27 under Model 1: TAA and TAG encode glutamine, and the context-dependent TGA is excluded. Under Model 2 its tryptophan contribution breaks the match. Tables 11 and 28 under Model 1 retain the original pool of the standard code; table 11 retains it under Model 2 as well.

Under Model 1 the pair of differences is reproduced by the standard table, five alternative nuclear tables and table 11, which is used by bacteria, archaea and plastids. None of the 13 mitochondrial tables reproduces it. The failure is not due to the reassignment of ATA or TGA alone: in tables 16, 22 and 23 both codons retain their standard assignments, whereas in the remaining ten at least one of them is changed. In table 16 TAG encodes leucine; in table 22 the conversion of TCA into a stop codon is added to this; in table 23 TTA becomes a stop codon. After these reassignments additional contributions enter the difference of the Keto and Amino sums, so that it no longer reduces to the difference of the nucleon counts of tryptophan and isoleucine.

Preservation of the differences and divisibility of the sums. In the standard service-excluded pool the neutron sums of Keto and Amino are divisible by 37. The proton sum of Amino is also divisible by 37, while the proton sum of Keto has residue -1 modulo 37. This set of residues is reproduced by exactly the same tables as the standard values of the neutron deficit and the proton residue in the first test: 1, 11 and 28 under Model 1 and 1, 11 under Model 2.

In tables 6, 27, 29 and 30 the symmetric reassignment of TAA and TAG contributes equally to both halves, separately for protons and for neutrons; these contributions are not divisible by 37. In the models where the standard differences are preserved, none of the four sums is divisible by 37: the neutron residues are equal and non-zero, and the proton residue of Keto is one less than the proton residue of Amino.

The localization of the asymmetry in the tryptophan–isoleucine pair is thus preserved under the symmetric reassignment of TAA and TAG as well. Equal increments to both halves, however, preserve their differences while changing the residues of the sums themselves modulo 37. Preservation of this local asymmetry therefore does not amount to reproducing the arithmetic regularities of the original pool of the standard code.

Comparison of the minimal assignments. For each table and both models the dataset `service_codon_balance_comparison.csv` lists two minimal assignments: under Keto/Amino balance alone and with the global divisibility by 37 added. In each procedure every stop codon of the adopted model receives the same value and ATG a separate one; the total added contribution is minimized under non-negative integer parameters, separately for P and N. The parameters are derived anew for each table.

Balance with divisibility has a joint solution in protons and neutrons not for every table: under Model 1 it is absent for tables 14, 25 and 33, and under Model 2 also for 27, 28 and 31. Under balance alone a joint solution is absent only for table 25. For tables 14 and 33 the minimal total contribution is determined separately for protons and for neutrons, but its distribution among the service codons is not uniquely determined.

Under balance alone the minimal neutron compensation leads to equality of the neutron sum of the complete table with the nucleon sum of its structural Octet I only in tables 1, 11 and 28

under Model 1 and 1, 11 under Model 2. Neither this equality nor divisibility by 37 is imposed in the procedure. For the standard code the result is 37 neutrons for each stop codon and a zero contribution from ATG; the sum equals $3589 + 3 \cdot 37 = 3700$.

With divisibility additionally required the result broadens: equality with the nucleon sum of structural Octet I also holds for tables 2, 9, 10, 13, 21 and 24 under both models. Among the tables with a joint solution of balance with divisibility this equality is attained in 9 of 24 tables under Model 1 and in 8 of 21 under Model 2. After the minimal assignment in both components, these six tables agree with the standard code under Key 1 in all four quantities across all 33 groups. The assignments themselves, however, differ. For table 2, for instance, the result is $(P, N) = (26, 24)$ per stop codon and $(80, 69)$ for ATG, and for table 10 $(10, 17)$ and $(28, 20)$ respectively. Table 10 differs from the standard table only in reading TGA as cysteine; in the other five, further amino acid assignments are changed as well.

For tables 12 and 26 the proton and neutron sums after this procedure also coincide with the standard values 4292 and 3700, but the full group profile differs. Agreement of the complete sums, including the nucleon sum $7992 = 8 \cdot 999$, therefore does not amount to agreement of the sums of the individual groups.

The same minimal assignment as for the standard code — $(37, 37)$ for each stop codon and $(1, 0)$ for the initiator ATG — is obtained only for tables 1, 11 and 28 under Model 1 and 1, 11 under Model 2. What is distinguished is the form of the assignment rather than the magnitude of the added contribution itself: tables 9, 10, 13, 21 and 24 reproduce the full profile with a smaller total added contribution in both P and N.

The additional contribution required to satisfy divisibility once the minimal balance has been reached amounts, for the standard code, to four protons and zero neutrons. Among the tables with a joint solution this is the smallest increment in each component separately. The same values are obtained for table 11 under both models and for table 28 under Model 1. For the remaining tables with a joint solution both increments are larger.

The standard values of the neutron deficit and the proton residue are thus reproduced only in tables with the same service-excluded pool. The localization of the asymmetry in the tryptophan–isoleucine pair is preserved under some changes of that pool as well. After the assignment to the service codons the agreements broaden: other codes can reproduce the entire group profile of the standard code, but at different parameter values. Agreement of the resulting sums therefore does not by itself amount to agreement of the underlying arithmetic or of the assignment to the service codons.

These conclusions concern the tables of the registry examined here. The number of matches does not provide an estimate of probability, since the tables are not an independent random sample of genetic codes.

4.2 Specificity of the third codon position, the modulus 37, and the molecular representation

The third codon position, the modulus 37 and the representation of free neutral amino acids define the framework of the analysis (Sections 2.3, 2.6 and 2.1). What follows tests which properties are preserved when these choices are changed. The positional control compares the divisibility of the sums along the same chemical axes; the prime scan, how widely divisibility occurs across different moduli; the representation control, the stability of the absolute sums and of the differences. An additional a posteriori observation on the Keto/Amino asymmetry across the three positions is then considered.

Codon position. We take the positional control over the service-excluded pool: its sums do not depend on the assignments to the service codons. The three chemical axes are applied to each position in turn. At the third position the Amino and Weak groups have all four quantities divisible by 37; the Keto and Strong groups have their neutron sums divisible by 37, and the

Pyrimidine group its difference Δ (Section 3.1.1). For Keto and Strong the proton and nucleon sums lie one unit below the nearest multiple of 37:

$$P_{\mathrm{SE}}(\mathrm{Keto}) = P_{\mathrm{SE}}(\mathrm{Strong}) = 2108 = 57 \cdot 37 - 1,$$
$$T_{\mathrm{SE}}(\mathrm{Keto}) = T_{\mathrm{SE}}(\mathrm{Strong}) = 3921 = 106 \cdot 37 - 1.$$

Since the neutron sums are divisible by 37, the identities $T = P + N$ and $\Delta = P - N$ carry the residue of the proton sum over to T and Δ. At the first position only the neutron sums of Keto and Amino are divisible. Given the divisibility of the total sum of the pool, $N(\mathcal{C}_{\mathrm{SE}}) = 3589 = 97 \cdot 37$, divisibility of one of these halves forces divisibility of the other, so the two matches are not independent. At the second position none of the four quantities of any of the six groups is divisible by 37.

The chemical axes split the full table into halves of 32 codons, but once the service positions are excluded the sizes may differ. In particular, Keto and Amino at the first position contain 29 and 31 codons, and at the third position 30 each. In the pool considered here the divisibilities of the group sums are thus concentrated at the third position; carrying the same chemical classifications over to the other positions does not reproduce this set of divisibilities. The values for all three positions in the service-excluded pool (identified in the file by the value `sense`), under Key 0 and under Key 1 are given in `position_axis_analysis.csv`.

Choice of modulus. The scan covers all primes up to 97 and three sets of group quantities: the values of T, P, N and Δ for those 17 groups of the 33 that contain no service codons; the values of all 33 groups under Key 0; and the values of the same groups under Key 1. In the file the sets are identified by the values `param_independent`, `key0` and `key1`. In the first set the assignments to the service codons do not enter the sums; in the second, divisibility by 37 is not imposed; and in the main derivation of Key 1 it enters the conditions on the full sums P and N.

For each prime p we count the share of the distinct non-zero values that it divides. Repeated numerical values are counted once. This share is compared with the value $1/p$, which corresponds to a uniform distribution of residues. In all three sets the largest ratio to this value belongs to 37: it divides 5 of the 28 distinct values in the first set, 13 of 70 under Key 0 and 10 of 47 under Key 1. The corresponding ratios are approximately 6.61, 6.87 and 7.87. For 2 and 5 they are below three in all three sets; divisibility by these primes is connected, respectively, with the parity of the proton counts and with decimal roundness, discussed in Sections 3.3.2 and 3.6.

The ratio to $1/p$ is a descriptive statistic: the scan specifies no null model of genetic codes and yields no p-value. If k of the m distinct values are divisible by p, the ratio equals pk/m: it does not exceed p and changes in steps of p/m. The range and the step of the statistic therefore depend on the size of the modulus. Removing duplicate values does not remove the dependencies induced by the addition and the scaling of the group sums. The range of comparison is limited to the primes up to 97. Powers of primes impose additional divisibility conditions and are not part of this comparison. It is established separately that no value in the three sets is divisible by 37^2.

In addition, each positive value is reduced to its odd core by removing all factors of 2. This eliminates repetitions of the form v, $2v$ and $4v$ that arise, in particular, from the exact scaling of the Octet I groups (Section 3.1.2). Under Key 0 the number 37 divides ten distinct odd cores, whereas the multiples of each of 73 and 61 reduce to a single core. For 73 this is the value itself, $73 = 2 \cdot 37 - 1$, from the Octet II Δ cascade; for 61 it is the value $549 = 9 \cdot 61$, to which the proton sums of the pyrimidine groups reduce: $2196 = 4 \cdot 549$ and $1098 = 2 \cdot 549$. This count does not establish the independence of the remaining values. The full counts are given in `prime_divisibility_scan.csv`. The first place of 37 under the chosen metric is already observed in the set for the groups without service codons and under Key 0; it is not due to the Key 1 assignments.

Dependence on the molecular representation. We compare the adopted representation of the free neutral amino acid with the peptide residue and with the zwitterion, keeping the isotope

convention of Section 2.1. In the peptide-residue model one water molecule, that is 10 protons and 8 neutrons, is subtracted from the formula of each amino acid. The zwitterionic form has the same atomic composition as the free neutral molecule, so its nucleon counts coincide with the original ones.

These changes are described by a homogeneous integer shift (s_P, s_N): the same pair of values is added to the counts of every amino acid. In the difference of two sums of one quantity the additions cancel if the number of summands is the same. The differences of the molecular counts of tryptophan and isoleucine are therefore preserved, $\delta N(\mathrm{Trp}, \mathrm{Ile}) = 37$ and $\delta P(\mathrm{Trp}, \mathrm{Ile}) = 36$. The differences of the Keto/Amino and Strong/Weak sums at the third position are preserved as well: in the service-excluded pool each of these groups contains 30 codons. This is stability with respect to homogeneous shifts; an arbitrary change of the molecular representation may change different amino acids differently.

In the absolute sums equal additions do not cancel. Divisibility of the neutron sum of the pool by 37 is preserved under the condition $60 s_N \equiv 0 \pmod{37}$, that is $s_N \equiv 0 \pmod{37}$, since 60 and 37 are coprime. For the peptide residues the neutron and proton sums are, respectively,

$$3589 - 60 \cdot 8 = 3109 = 84 \cdot 37 + 1,$$
$$4180 - 60 \cdot 10 = 3580 = 97 \cdot 37 - 9.$$

Both lose their original residues modulo 37. The neutron sums of Keto and Amino at the third position become 1573 and 1536: the difference 37 is preserved, but neither sum is a multiple of 37 any longer. The nucleon sum of Octet I in the same representation equals $3700 - 32 \cdot 18 = 3124$, so its difference from the neutron sum of the pool is $3124 - 3109 = 15$ instead of the original 111.

The absolute arithmetic properties thus belong to the adopted molecular representation. The differences between groups of the pool of equal cardinality and between molecules are preserved under the homogeneous shifts tested here, but the stability of the whole structure does not follow from this. All 13 quantities tested under the three representations are given in `representation_sensitivity.csv`.

The Keto/Amino asymmetry across the three positions. We now compare the differences of the Keto minus Amino sums in the service-excluded pool. In the standard code, at each position exactly one of the four differences in T, P, N and Δ is divisible by 37:

Position	1	2	3
Quantity	N	T	N
Difference	$-333 = -9 \cdot 37$	$185 = 5 \cdot 37$	37

At the second position the divisibility arises in the subtraction: the nucleon sums themselves are not multiples of 37, but they have equal residues, which cancel in the difference.

At the first two positions the contributions of all nineteen amino acids of the pool remain in the differences (methionine drops out together with ATG). At the third position synonymy cancels the contributions of every amino acid except tryptophan and isoleucine. The neutron difference of the two groups thereby reduces to the difference of a single pair of molecules (Section 3.3.1):

$$N_{\mathrm{SE}}(\mathrm{Keto}) - N_{\mathrm{SE}}(\mathrm{Amino}) = N(\mathrm{Trp}) - N(\mathrm{Ile}) = 37.$$

The comparison of the 27 NCBI tables shows that the combination of divisibilities in N, T, N across the three positions is reproduced only in tables 1, 11 and 28 under Model 1 and in tables 1 and 11 under Model 2. In all these cases the composition of the analyzed pool and its amino acid assignments coincide with the standard ones. For table 28 this agreement is obtained when the context-dependent stop codons are excluded under Model 1; Model 2 returns them to the pool (Section 4.1). In the remaining tables the divisibility for this sequence is preserved at no more than one position. The full results are given in `position_asymmetry_ncbi.csv`.

The sequence N, T, N was singled out after inspection of the data. The comparison of the tables describes how widely this combination occurs in the registry and gives no estimate of the probability of its arising by chance.

All four control datasets are computed deterministically in exact integer arithmetic by the script `controls.py` (Section 2.7).

4.3 Significance under a random-code null

The comparison with natural codes and the deterministic controls (Sections 4.1 and 4.2) describe the specificity of the observed structure. Comparing natural codes does not by itself determine the probability of the observed properties under random permutations of amino acids. Here we estimate how often randomly reassigning amino acids to the blocks of the standard code produces a comparable concentration of group quantities at and near multiples of 37. Comparisons with random alternatives are used in studies of genetic-code optimality [2].

The null model preserves the 64 codons, the twenty synonymous blocks with their sizes, the three stop codons, the Rumer partition, and the third-position chemical axes. In each trial, the twenty real amino acids are uniformly randomly permuted among these blocks; each amino acid's proton and neutron counts are transferred together. Six main summaries are considered. `S1`–`S3` and `S4sense` are evaluated in the service-excluded pool (60 codons): ATG, TAA, TAG, and TGA are excluded from every group in every trial. `S4a` and `S4b` extend the analysis to the complete table under Key 0 and Key 1, respectively.

For each of the six main summaries, $M = 10^6$ trials give the estimated tail probability $p_{\mathrm{MC}} = (b+1)/(M+1)$, where b is the number of random codes with a result at least as extreme as the observed one. Equal or greater values are counted for divisibility counts, and equal or smaller values for mean distances. Each estimate is accompanied by a 95% Wilson interval for the proportion b/M, describing Monte Carlo uncertainty. The calculation is implemented in `mc_significance.py` with the random seed `seed=0`; the summary identifiers and results are reported in `mc_significance.csv`. The independent implementation `verify_mc.py` uses a different construction of the codon groups, a direct search when deriving Key 1, and a different random seed (`seed=12345`). It reproduces the observed statistics exactly and the tail probabilities within the expected Monte Carlo error.

`S1` tests joint divisibility by 37 of the neutron sum of the service-excluded pool and the nucleon sum of Octet I. In the standard code these are $N(\mathcal{C}_{\mathrm{SE}}) = 3589 = 97 \cdot 37$ and $T(\text{Octet I}) = 3700 = 100 \cdot 37$. In random codes, both sums are tested for divisibility without requiring them to match their exact values in the standard code. The result is $p_{\mathrm{MC}} = 0.000800$ with a 95% interval of 0.000746–0.000856. The joint event is estimated directly, without assuming independence of the two divisibilities.

`S2` counts quantities divisible by 37 among T, P, N, and Δ for nine groups restricted to the service-excluded pool: the full pool, the six chemical-axis half-pools, and the two octets. In the standard code, 13 of 36 quantities are divisible, whereas the mean in random codes is 1.0299. A result of at least 13 occurred in 39 of 10^6 trials: $p_{\mathrm{MC}} = 4.0 \times 10^{-5}$, 95% interval 2.9×10^{-5}–5.3×10^{-5}. The quantities are linked by the identities $T = P + N$ and $\Delta = P - N$ and by pairwise equality of the sums of the twin groups Keto/Strong and Amino/Weak; sums of complementary groups add up to the total for the full pool. These dependencies are preserved in every permutation and are already accounted for in the comparison with random codes.

Given these identities, the observed 13 divisibilities reduce to divisibility by 37 of five quantities: $N(\mathcal{C}_{\mathrm{SE}})$, $N_{\mathrm{SE}}(\text{Keto})$, $P_{\mathrm{SE}}(\text{Amino})$, $\Delta(\text{Pyrimidine})$, and $T(\text{Octet I})$. These are five linearly independent congruences modulo 37; statistical independence of the events is not assumed. The `S2` test includes any set of 13 or more divisibilities, including sets that differ from the observed one.

`S3` applies the same count only to the 17 groups that originally contain no service codons, giving 68 quantities in total. The standard code has 14 divisible quantities, against a mean of

2.1311 in random codes: $p_{\mathrm{MC}} = 0.025135$, 95% interval 0.024829–0.025443. Eleven divisibilities follow from $T(\text{Octet I}) = 100 \cdot 37$, and the remaining three form the pyrimidine Δ-cascade: $\Delta(\text{Pyrimidine}) = 8 \cdot 37$ and $\Delta(\{\text{C}\}) = \Delta(\{\text{T}\}) = 4 \cdot 37$. This secondary summary thus records the propagation of two structural cascades.

`S4sense` measures the overall proximity of group quantities to multiples of 37. For each of the 33 groups restricted to the service-excluded pool, the distances of T, P, N, and Δ to the nearest multiple of 37 are calculated; all 132 distances are then averaged. Zero values are included with a distance of zero. The mean distance is 4.8182 in the standard code and 9.1495 in random codes: $p_{\mathrm{MC}} = 0.000274$, 95% interval 0.000242–0.000307. The pool's atypicality is therefore also apparent in proximity to multiples, including values that are not exactly divisible by 37.

In an additional post hoc check (`S4pos`), zero distances were excluded from the `S4sense` distance set in the standard code and in each random code. If all distances are zero, the statistic is defined as zero; such trials are not discarded. The mean of the remaining distances is 6.1154 in the standard code and 9.4195 in random codes: nominal $p_{\mathrm{MC}} = 0.001602$, 95% interval 0.001525–0.001681. Thus, the deviation from random codes persists after exact hits on multiples of 37 are excluded.

In the main model, excluding ATG removes whichever amino acid was assigned to its block. To examine the role of this choice, an additional conditional model fixes methionine at ATG and permutes the other 19 amino acids among the remaining blocks. The results are $p_{\mathrm{MC}} = 0.000026$ for `S2`, 0.000268 for `S4sense`, and 0.001531 for `S4pos`. Atypicality persists when the set of amino acids represented in the pool is held fixed.

The additional calculations use 10^6 trials per model in `mc_sensitivity.py` (`seed=0`); full estimates and intervals are provided in `mc_sensitivity.csv`. The independent implementation `verify_mc_sensitivity.py` (`seed=12345`) reproduces the observed values exactly and the means and tail probabilities within the expected Monte Carlo error.

`S4a` calculates the same mean distance under Key 0. Stop codons contribute zero, and ATG contributes the profile of the amino acid assigned to its block in the code being evaluated: methionine in the standard code and the result of the permutation in a random code. The mean distance is 5.5606 in the standard code and 9.0157 in random codes: $p_{\mathrm{MC}} = 0.002807$, 95% interval 0.002704–0.002912. Proximity to multiples is thus retained when service codons are restored under this formal assignment.

`S4b` evaluates the mean distance after deriving Key 1. For the standard code and each random code, a separate minimal non-negative assignment is derived from the Keto/Amino balance and divisibility of the global proton and neutron sums by 37 (Section 3.4.3). Divisibilities imposed by the procedure itself therefore also arise in random codes. The mean distance is 4.6364 in the standard code and 7.7397 in random codes: $p_{\mathrm{MC}} = 0.021106$, 95% interval 0.020825–0.021389. This probability describes the proximity of group quantities to multiples of 37 after the same key-derivation procedure has been applied to every code being compared.

Under the chosen model of random amino acid reassignment, the standard code stands out primarily through the properties of its service-excluded pool. More group quantities are exactly divisible by 37, and the mean distance to the nearest multiple is smaller than is typical of random codes. After exact hits are excluded, the remaining quantities are also closer to multiples on average. Thus, the observed proximity is not exhausted by exact divisibilities alone. The divisibility count and the overall mean distance are linked: exact divisibilities enter this mean with zero distance. These results describe different aspects of the same arithmetic structure. Fixing methionine at ATG also preserves this atypicality. Proximity to multiples persists when service codons are restored under the two formal assignments; after Key 1 is re-derived, the deviation from random codes is less pronounced.

All reported p_{MC} values are nominal probabilities for individual summaries: no multiplicity correction was applied to either the main summaries or the additional checks, and dependent p-values are not combined by multiplication. The modulus 37 is taken from prior literature; the

main summaries were formally defined before the final Monte Carlo run but after inspection of the observed structure. The analysis was not preregistered and is exploratory. The resulting probabilities refer to the models considered here, with the standard code's blocks fixed and the amino acids' real nucleon profiles. They do not estimate the probability of a natural code arising or identify a biological mechanism.

4.4 Relation to prior work

Previous work by Rumer, Shcherbak and Makukov, and Panov and Filatov is particularly relevant to the arithmetic structure analyzed in the present study. Yu. B. Rumer [6, 7] introduced a partition of the 64 codons of the standard code into two equal classes: Octet I, consisting of eight fully degenerate families, and Octet II, consisting of eight heterogeneous families. These two classes are related by the involution $R = (\mathrm{T} \leftrightarrow \mathrm{G}, \mathrm{C} \leftrightarrow \mathrm{A})$, which maps each family in one class to a family in the other. Shcherbak and Makukov [8] noted the recurring role of 37 in the genetic code as a divisor of amino acid nucleon sums across different codon groupings. Panov and Filatov [5] revisited and extended this analysis. In particular, they considered octet sums under per-codon counting, where each codon contributes the total nucleon count of the amino acid it encodes, and stop codons are assigned zero. They also proposed an alternative assignment of a total nucleon count of 74 to each stop codon.

The present work differs from these studies in several methodological choices. Alongside whole molecules, Shcherbak and Makukov, and subsequently Panov and Filatov, consider the decomposition of an amino acid into a standard block and a variable side chain, and many of their nucleon sums involve side chains alone. Making the blocks uniform requires special treatment of proline, whose cyclic side chain reduces the nucleon count of its block to 73 rather than 74: to bring it into line with the other nineteen amino acids, Shcherbak and Makukov introduce a virtual transfer of one nucleon from its side chain to its block — an operation they call the *activation key*. The total nucleon count of proline is unchanged: $42 + 73 = 41 + 74 = 115$. The present work uses the total nucleon counts of amino acids without separating blocks and side chains, so the question of this redistribution for proline does not arise. Both previous studies use per-codon counting as well as contracted counting, in which each amino acid is counted once in each family that encodes it, regardless of the number of corresponding codons in that family. The present work consistently uses per-codon counting. Finally, the results of previous work considered below concern the total nucleon count T or its decomposition into blocks and side chains. Our analysis separately considers protons P and neutrons N, as well as the difference $\Delta = P - N$; this allows us to investigate relations that are not determined by the sum T alone.

The two previous studies also differ from ours in the schemes used to group codons. They employ several grouping schemes in parallel (Gamow's sorting and others), within which different arithmetic regularities are considered. The present work formulates a single grouping system based on Rumer's octets and the chemical axes of the third position, and consistently analyzes the arithmetic structure within it. The parametrization of service codons also differs: Shcherbak and Makukov treat stop codons as contributing zero and count ATG as encoding methionine; Panov and Filatov additionally propose assigning a total nucleon count of 74 to stop codons, whereas the present work treats the parametrization of service codons as part of the derivation of Key 1 (Section 3.4). The same stop-codon contribution of 74 does not mean that the assignments coincide: Panov and Filatov retain the methionine contribution of ATG, equal to 149, whereas under Key 1 $T_1(\mathrm{ATG}) = 1$; our key also specifies proton and neutron contributions separately. Finally, alongside the standard code, Shcherbak and Makukov consider the euplotid variant (in which TGA is reassigned to cysteine) as a symmetrized counterpart of the standard code; Panov and Filatov also consider this reassignment. The present work focuses on the standard code and compares its arithmetic structure with 26 alternative NCBI translation tables under common counting rules (Section 4.1).

We now describe how divisibility results from these previous studies map onto the groups

defined in the present analysis. Under contracted counting, where each amino acid is counted once in each family that encodes it, Shcherbak and Makukov obtained the sum $925 = 25 \cdot 37$ for Octet I. For Octet II, with stop codons contributing zero, their balance between blocks and side chains implies the sum $1110 + 1110 = 2220 = 60 \cdot 37$. These sums refer to the contracted representation, not to summation over all codons in the octet. Using per-codon counting, Panov and Filatov obtained sums corresponding to our Key 0: $T(\text{Octet I}) = 3700 = 100 \cdot 37$ and $T_0(\text{Octet II}) = 4218 = 114 \cdot 37$. Under their alternative assignment of 74 to each stop codon, the second sum becomes $4440 = 120 \cdot 37$.

The relationship between the contracted sum and the per-codon sum is direct for Octet I, which is fully four-fold degenerate: $3700 = 4 \cdot 925$. For Octet II, whose coding series have different multiplicities, the per-codon sum 4218 cannot be obtained by uniformly scaling the contracted sum 2220: each contribution must be counted with its own codon multiplicity. The contracted sum 2220 itself coincides with $T_0(\text{Keto} \cap \text{Octet II}) = T_0(\text{Strong} \cap \text{Octet II})$ in our analysis. In the standard code, each of these sixteen-codon halves of Octet II contains one codon for each amino acid of its respective family, while stop codons contribute zero. Its per-codon sum therefore equals the contracted sum. The equality of the Keto and Strong halves follows from pyrimidine synonymy (Section 2.4); the agreement with contracted counting additionally uses the actual distribution of products within the families.

These facts imply divisibility by 37 for several further groups in our system under Key 0. The sum for the full code is the sum of the two octet sums; the sixteen-codon chemical-axis groups within Octet I have equal sums of 1850, and the eight-codon groups with a fixed third base have sums of 925, owing to four-fold degeneracy. Each of the four groups Keto, Strong, Amino and Weak splits into a sixteen-codon part of Octet I and a sixteen-codon part of Octet II. For Keto and Strong, the sum of the second part is 2220, and for Amino and Weak it is $4218 - 2220 = 1998 = 54 \cdot 37$. Thus, divisibility of the total sums T of these four groups under Key 0 already follows from the equalities above; their proton and neutron sums require separate analysis.

4.5 Limitations and outlook

The results characterize the arithmetic structure of the genetic code table under explicitly specified counting rules. Each codon is counted once, so the weight of an amino acid is set by the number of codons encoding it in the group considered. Such sums do not describe the composition of a particular protein or cell, where codon usage frequencies matter. The service-excluded pool is likewise an analytical object: excluding ATG together with the three stop codons separates the results from the choice of service-codon assignments, but does not mean that ATG lacks its function of encoding methionine.

The nucleon quantities refer to the adopted molecular and isotopic representation (Section 2.1). Isotopic substitutions that leave the elemental composition unchanged alter N and T while preserving P. In the transition to peptide residues, some divisibilities of the absolute sums are lost, whereas differences between groups of equal cardinality are preserved under homogeneous shifts of the molecular quantities (Section 4.2). The established stability of these differences therefore does not imply that the whole arithmetic structure is independent of the representation.

The statistical comparison assesses the placement of amino acids within a fixed code architecture: in the main model the twenty amino acids are permuted among the synonymous blocks, and in the additional model methionine is fixed at ATG (Section 4.3). These models do not describe the evolutionary origin of the architecture itself. The modulus 37 is motivated by prior literature, but the summaries were chosen after inspection of the observed structure. Their nominal p-values do not estimate the probability of finding an arbitrary arithmetic regularity after a search over groups, moduli, and metrics. Dependencies between quantities are preserved in every permutation and are accounted for in the comparison, but recurring relations receive additional weight in the summaries. The robustness of the conclusions to changes in these weights and to other null

models remains a separate question.

The uniqueness of Key 1 refers to the minimum under the adopted conditions: equal assignments to the three stop codons, non-negative integer parameters, Keto/Amino balance, and separate divisibility of the total sums P and N by 37. The same minimum is also given by other additional conditions, in particular by an additional balance in Δ (Section 3.4.3). The coincidence of the results follows from the properties of the service-excluded pool and is not an independent confirmation. The uniqueness of the minimum does not imply a unique choice of conditions. The resulting assignments have a formal arithmetic meaning and are not interpreted as the composition of actual products of the service codons. Equalities imposed by the conditions of the derivation, or following from them, likewise do not serve as independent evidence for these conditions. The equality $N_1(\text{All}) = T(\text{Octet I})$ also depends on the premise of equal assignments to the stop codons: without it, with the other conditions retained, the minimal neutron sum of the complete table is 3626, and the assignments to individual service codons are not unique.

The specificity among the 27 NCBI tables is limited to the registry examined and to the rules for treating context-dependent stop codons. The tables do not form an independent random sample, so the number of matches has no probabilistic interpretation. After the assignment to the service codons is re-derived, different codes can reproduce the same profile of all 33 groups (Section 4.1). This profile therefore does not determine the code uniquely; agreement of the resulting sums must be distinguished from agreement of the initial pool and of the assignments themselves.

The immediate directions for further work are tests of sensitivity to biochemical constraints on the permutations and to group weights, with both choices justified a priori, and a description of the classes of codes with identical arithmetic profiles. The search for biological mechanisms that could link the observed divisibilities and equalities to the functioning or origin of the code is left for future research. Such hypotheses should yield testable predictions beyond the numerical relations already established.

5 Conclusion

The regularities examined here link the chemical organization of the codon table to the nucleon composition of the encoded amino acids. Partitioning by the Rumer octets and by the properties of the third base gives rise to mutually consistent divisibilities, equalities, and one-unit deviations from multiples of 37. Their values depend jointly on the composition of the amino acids and on which codons correspond to them: each amino acid enters a group sum with the multiplicity of its codons. Counting protons and neutrons separately brings out relations that are not apparent when only nucleon sums are considered.

The principal agreement is present already in the service-excluded pool. Its neutron sum is $3589 = 97 \cdot 37$, and its proton sum is $4180 = 113 \cdot 37 - 1$. The same pattern of deviation appears in the difference between Keto and Amino: after mutual cancellation of contributions, what remains is the difference of a single pair, tryptophan–isoleucine, equal to 37 in neutrons and 36 in protons. The agreement of the local differences with the global sums makes both the Keto and the Amino neutron sums divisible by 37, while the one-unit proton deviation is concentrated in Keto. The parity of the proton counts of the amino acids rules out a difference of 37, but the value 36 itself is determined by the specific molecular composition of this pair.

Another link between the levels of the partition is shown by the difference $\Delta = P - N$. In the eight-codon groups of Octet I it equals $75 = 2 \cdot 37 + 1$, and in the corresponding pyrimidine groups of Octet II it equals $73 = 2 \cdot 37 - 1$. When the octets are combined, the deviations cancel: $75 + 73 = 148 = 4 \cdot 37$. The relation $75 + 73 = 148$ recurs for the codon groups with C and T in the third position, because such codons pairwise encode the same amino acids. Under Key 0 a further agreement arises: counting methionine with its $149 = 4 \cdot 37 + 1$ nucleons compensates for the remainder of -1 in the nucleon sum of the pool. Thus, a number of exact divisibilities are

obtained before Key 1 is derived.

Key 1 links the initial asymmetry to the placement of the service codons. The Keto group contains one stop codon and ATG, and the Amino group contains two stop codons. With the same assignment to all stop codons, Keto/Amino balance in protons and neutrons, and divisibility of the total sums P and N by 37, the unique minimal non-negative integer assignment is $(P, N) = (37, 37)$ for each stop codon and $(1, 0)$ for ATG. The neutron equality is especially instructive:

$$N_1(\text{All}) = 3589 + 3 \cdot 37 = 3700 = T(\text{Octet I}).$$

The size of the local imbalance and the number of stop codons give exactly the increment that brings the neutron sum of the complete table to the nucleon sum of Octet I. This coincidence with Octet I was not used as a condition in deriving the assignment.

The resulting assignment also brings other partitions into agreement. Keto/Amino balance, together with the twin-group identities, ensures that each of the four quantities is equal across the Keto, Amino, Strong, and Weak groups. The difference $\Delta = 296 = 8 \cdot 37$ takes one and the same value for all six chemical-axis halves, including Purine and Pyrimidine, whose separate sums P and N remain different. The link with Octet I can also be written equivalently as $P_1(\text{All}) = T_1(\text{Octet II}) = 4292$; the nucleon sum of the complete table equals $7992 = 8 \cdot 999$, although divisibility by 999 was not separately required. These equalities form a system of dependent consequences and additional numerical agreements. Its content is therefore determined by the relations between the quantities, and the number of recurring equalities is not a count of independent pieces of evidence.

The checks delimit the domain in which this structure stands out. In the grouping system considered, the third position differs from the first two, and the modulus 37 ranks first among the tested primes under the chosen descriptive metric, already without the Key 1 assignments. The absolute divisibilities, however, depend on the molecular representation. The comparison with the NCBI tables requires distinguishing the initial arithmetic of the pool from the profile after the assignment to the service codons: some alternative codes reproduce the entire resulting group profile of the standard code with other assignments. This profile alone therefore does not determine the standard code uniquely.

Permutation tests show that, with the synonymous blocks preserved, random redistribution of the amino acids rarely produces an equally high concentration of group quantities near multiples of 37. The deviation is most pronounced in the service-excluded pool: for the number of divisibilities and the mean distance, the nominal values are $p_{\text{MC}} = 4.0 \times 10^{-5}$ and 2.74×10^{-4}, respectively. Proximity to multiples persists after exact hits are excluded, and fixing methionine at ATG does not remove the atypicality either. These dependent checks characterize different aspects of the observed agreement under the adopted models; they do not give a probability that the code arose by chance.

Thus, the work describes how the specific correspondence between codons and amino acids links the local asymmetry, the sums of large groups, and the conditions of a formal assignment to the service codons. The initial relations are established without such an assignment, and Key 1 shows that they can be reconciled in the complete table under explicitly stated conditions. The search for biological mechanisms that could explain these links remains a direction for further research.

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